Material Characterization Report

NanoVNA UTN Toolkit

2026-10-24 12:54:39

Measurement Details:

- **Sample:** Muestra13
- **Technique:** Open-ended coaxial probe (liquids)
- **Temperature:** 23.0 °C
- **Reference Liquids:** Distilled Water / Isopropyl Alcohol (IPA)
- **Probe:** 3 mm probe
- **Frequency Range:** 0.005–6.000 GHz, 1200 pts
- **Data sources:**
  - **Open:** kit:cal_sonda3mm_UBA_1200p
  - **Short:** kit:cal_sonda3mm_UBA_1200p (1 spike(s) replaced in the computation: 4885 MHz)
  - **Ref 1:** kit:cal_sonda3mm_UBA_1200p
  - **Ref 2:** kit:cal_sonda3mm_UBA_1200p
  - **DUT:** imported

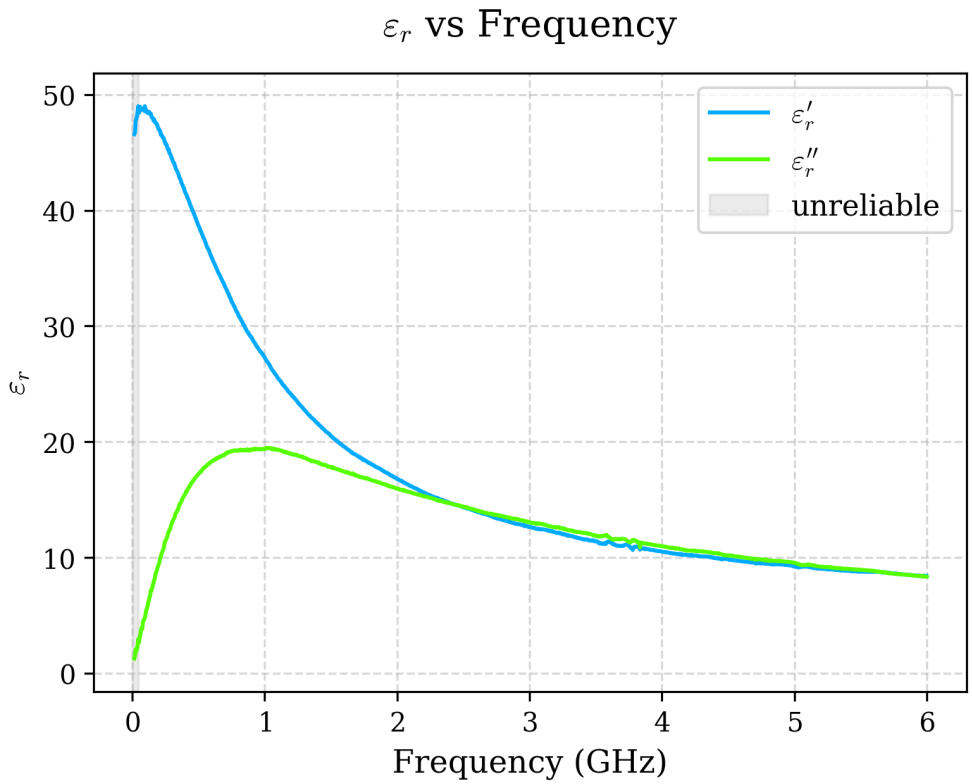
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1 Measurement Results

1.1 Complex Permittivity $\varepsilon_r(f)$



Points without a solution: 2 of 1200 (left blank in the plot and shown as — in the table).

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.005	—	—	—
0.010	—	—	—
0.015	46.5751	1.2631	0.0271
0.020	46.6837	1.4994	0.0321
0.025	47.6898	2.0763	0.0435
0.030	47.9039	1.9053	0.0398
0.035	48.0959	2.0984	0.0436
0.040	48.5196	2.4318	0.0501
0.045	48.9919	2.9709	0.0606
0.050	48.4280	2.6240	0.0542
0.055	48.4981	3.0544	0.0630
0.060	48.9399	3.2938	0.0673
0.065	48.6974	3.7086	0.0762
0.070	48.7634	3.9180	0.0803
0.075	48.7485	3.8262	0.0785
0.080	48.6946	4.4251	0.0909
0.085	48.6572	4.6248	0.0950

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.090	48.5874	4.6301	0.0953
0.095	48.9947	4.7873	0.0977
0.100	48.7517	5.0850	0.1043
0.105	48.4834	5.4678	0.1128
0.110	48.5619	5.5589	0.1145
0.115	48.4728	5.8443	0.1206
0.120	48.3787	6.1044	0.1262
0.125	48.5187	6.2563	0.1289
0.130	48.3877	6.5791	0.1360
0.135	48.3986	6.7178	0.1388
0.140	48.3350	7.1809	0.1486
0.145	48.0860	7.1519	0.1487
0.150	48.0016	7.3422	0.1530
0.155	47.9815	7.6130	0.1587
0.160	47.7481	7.8240	0.1639
0.165	47.8639	8.0708	0.1686
0.170	47.6636	8.1813	0.1716
0.175	47.5634	8.5866	0.1805
0.180	47.5257	8.7158	0.1834
0.185	47.4678	8.8594	0.1866
0.190	47.2642	9.1216	0.1930
0.195	47.1180	9.2696	0.1967
0.200	46.9725	9.4548	0.2013
0.205	46.9597	9.5773	0.2039
0.210	46.8483	9.9412	0.2122
0.215	46.5330	10.0618	0.2162
0.220	46.5439	10.3026	0.2214
0.225	46.4148	10.4476	0.2251
0.230	46.2948	10.5697	0.2283
0.235	46.2754	10.7507	0.2323
0.240	46.0933	10.9883	0.2384
0.245	45.9450	11.3448	0.2469
0.250	45.7334	11.4444	0.2502
0.255	45.7456	11.5277	0.2520
0.260	45.5944	11.6876	0.2563
0.265	45.4122	11.8384	0.2607
0.270	45.2226	12.0687	0.2669
0.275	45.2590	12.2586	0.2709
0.280	44.9489	12.3483	0.2747
0.285	44.8990	12.4456	0.2772
0.290	44.7268	12.6664	0.2832
0.295	44.5903	12.8476	0.2881
0.300	44.4620	12.9842	0.2920
0.305	44.2772	13.1732	0.2975
0.310	44.1203	13.2677	0.3007
0.315	44.1056	13.3872	0.3035

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.320	43.9356	13.4983	0.3072
0.325	43.7295	13.7603	0.3147
0.330	43.5471	13.8392	0.3178
0.335	43.4688	13.9383	0.3207
0.340	43.3199	14.0818	0.3251
0.345	43.2349	14.1493	0.3273
0.350	43.0398	14.3612	0.3337
0.355	42.8618	14.5611	0.3397
0.360	42.7318	14.6191	0.3421
0.365	42.6289	14.7455	0.3459
0.370	42.4066	14.8117	0.3493
0.375	42.2772	14.9970	0.3547
0.380	42.1566	15.1063	0.3583
0.385	42.0235	15.2354	0.3625
0.390	41.9089	15.3383	0.3660
0.395	41.6765	15.3869	0.3692
0.400	41.5365	15.5069	0.3733
0.405	41.4095	15.6829	0.3787
0.410	41.3218	15.6577	0.3789
0.415	41.1314	15.8313	0.3849
0.420	41.0004	15.9306	0.3885
0.425	40.8832	16.0119	0.3916
0.430	40.7414	16.1475	0.3963
0.435	40.5192	16.1712	0.3991
0.440	40.4132	16.3140	0.4037
0.445	40.2744	16.4581	0.4086
0.450	40.1263	16.5125	0.4115
0.455	39.9554	16.5193	0.4134
0.460	39.8676	16.6595	0.4179
0.465	39.7135	16.7227	0.4211
0.470	39.5626	16.8225	0.4252
0.475	39.3812	16.9030	0.4292
0.480	39.2998	16.9326	0.4309
0.485	39.1103	17.0123	0.4350
0.490	38.9533	17.0991	0.4390
0.495	38.7787	17.1589	0.4425
0.500	38.6512	17.1987	0.4450
0.505	38.5506	17.2846	0.4484
0.510	38.4047	17.3371	0.4514
0.515	38.2222	17.4238	0.4559
0.520	38.1434	17.4199	0.4567
0.525	37.9857	17.5233	0.4613
0.530	37.8377	17.6242	0.4658
0.535	37.6886	17.6866	0.4693
0.540	37.5715	17.7477	0.4724
0.545	37.4492	17.8081	0.4755

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.550	37.2761	17.7951	0.4774
0.555	37.1465	17.8728	0.4811
0.560	36.9462	17.9271	0.4852
0.565	36.8366	17.9617	0.4876
0.570	36.7028	18.0520	0.4918
0.575	36.6328	18.0827	0.4936
0.580	36.4930	18.1480	0.4973
0.585	36.3046	18.1525	0.5000
0.590	36.1820	18.1994	0.5030
0.595	36.0374	18.2747	0.5071
0.600	35.9361	18.3102	0.5095
0.605	35.7780	18.3210	0.5121
0.610	35.6207	18.3756	0.5159
0.615	35.5122	18.4211	0.5187
0.620	35.4053	18.4217	0.5203
0.625	35.2496	18.4835	0.5244
0.630	35.1510	18.4916	0.5261
0.635	34.9882	18.5788	0.5310
0.640	34.8726	18.5796	0.5328
0.645	34.7204	18.5814	0.5352
0.650	34.5901	18.6438	0.5390
0.655	34.5022	18.6878	0.5416
0.660	34.3729	18.6778	0.5434
0.665	34.2931	18.7171	0.5458
0.670	34.1331	18.7757	0.5501
0.675	34.0341	18.7446	0.5508
0.680	33.8903	18.8087	0.5550
0.685	33.7903	18.8245	0.5571
0.690	33.7022	18.8183	0.5584
0.695	33.5580	18.8603	0.5620
0.700	33.4163	18.9120	0.5660
0.705	33.3039	18.9986	0.5705
0.710	33.1812	18.9765	0.5719
0.715	33.0513	19.0302	0.5758
0.720	32.9249	19.1099	0.5804
0.725	32.8040	19.0855	0.5818
0.730	32.6904	19.0749	0.5835
0.735	32.4992	19.1208	0.5883
0.740	32.4256	19.1637	0.5910
0.745	32.3059	19.1949	0.5942
0.750	32.1118	19.1859	0.5975
0.755	32.0131	19.1873	0.5994
0.760	31.9237	19.2314	0.6024
0.765	31.8008	19.2331	0.6048
0.770	31.6722	19.2451	0.6076
0.775	31.6060	19.2340	0.6086

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.780	31.4234	19.2674	0.6132
0.785	31.3014	19.2352	0.6145
0.790	31.2380	19.2164	0.6152
0.795	31.1116	19.2464	0.6186
0.800	30.9778	19.2336	0.6209
0.805	30.8370	19.2843	0.6254
0.810	30.7493	19.2680	0.6266
0.815	30.6725	19.3213	0.6299
0.820	30.5577	19.3206	0.6323
0.825	30.4334	19.2501	0.6325
0.830	30.3679	19.2978	0.6355
0.835	30.2206	19.2888	0.6383
0.840	30.0971	19.2503	0.6396
0.845	30.0501	19.2699	0.6413
0.850	29.9101	19.2593	0.6439
0.855	29.8044	19.3242	0.6484
0.860	29.6899	19.2448	0.6482
0.865	29.5631	19.3183	0.6535
0.870	29.4921	19.3405	0.6558
0.875	29.4652	19.2814	0.6544
0.880	29.3303	19.2377	0.6559
0.885	29.2144	19.2635	0.6594
0.890	29.1746	19.3252	0.6624
0.895	29.0512	19.2716	0.6634
0.900	29.0390	19.2809	0.6640
0.905	28.9233	19.3557	0.6692
0.910	28.8134	19.3617	0.6720
0.915	28.6665	19.3333	0.6744
0.920	28.6153	19.3146	0.6750
0.925	28.5396	19.4071	0.6800
0.930	28.4525	19.3747	0.6809
0.935	28.3089	19.3585	0.6838
0.940	28.2498	19.3926	0.6865
0.945	28.1416	19.3548	0.6878
0.950	28.0895	19.3291	0.6881
0.955	28.0010	19.3640	0.6915
0.960	27.9062	19.3423	0.6931
0.965	27.8244	19.3869	0.6968
0.970	27.7795	19.3637	0.6971
0.975	27.6888	19.3314	0.6982
0.980	27.6371	19.3761	0.7011
0.985	27.5686	19.3729	0.7027
0.990	27.4431	19.3770	0.7061
0.995	27.3807	19.3975	0.7084
1.000	27.2889	19.3643	0.7096
1.005	27.1715	19.4222	0.7148

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.010	27.0920	19.4717	0.7187
1.015	26.9756	19.4191	0.7199
1.020	26.8990	19.4528	0.7232
1.025	26.8490	19.4678	0.7251
1.030	26.7327	19.4470	0.7275
1.035	26.6609	19.4523	0.7296
1.040	26.5511	19.4489	0.7325
1.045	26.4375	19.4360	0.7352
1.050	26.3383	19.4349	0.7379
1.055	26.2261	19.3645	0.7384
1.060	26.1619	19.4180	0.7422
1.065	26.0999	19.3317	0.7407
1.070	26.0267	19.3790	0.7446
1.075	25.8966	19.3788	0.7483
1.080	25.8442	19.3407	0.7484
1.085	25.7764	19.3126	0.7492
1.090	25.6901	19.3398	0.7528
1.095	25.5828	19.3263	0.7554
1.100	25.5124	19.3207	0.7573
1.105	25.4369	19.2863	0.7582
1.110	25.3438	19.2770	0.7606
1.115	25.2807	19.2724	0.7623
1.120	25.2356	19.2630	0.7633
1.125	25.1565	19.2316	0.7645
1.130	25.0492	19.2224	0.7674
1.135	24.9500	19.1785	0.7687
1.140	24.9112	19.1159	0.7674
1.145	24.8290	19.1605	0.7717
1.150	24.8089	19.1158	0.7705
1.155	24.6972	19.1366	0.7748
1.160	24.6296	19.0712	0.7743
1.165	24.5565	19.1000	0.7778
1.170	24.4960	19.0644	0.7783
1.175	24.3808	19.0218	0.7802
1.180	24.3274	19.0517	0.7831
1.185	24.2910	19.0052	0.7824
1.190	24.2068	18.9610	0.7833
1.195	24.1380	18.9789	0.7863
1.200	24.0880	18.9835	0.7881
1.205	23.9906	18.8991	0.7878
1.210	23.9541	18.9353	0.7905
1.215	23.8628	18.9138	0.7926
1.220	23.8396	18.8950	0.7926
1.225	23.7767	18.8885	0.7944
1.230	23.6977	18.8996	0.7975
1.235	23.5995	18.8434	0.7985

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.240	23.5673	18.8633	0.8004
1.245	23.5033	18.8443	0.8018
1.250	23.3920	18.7984	0.8036
1.255	23.3240	18.7700	0.8047
1.260	23.2911	18.7646	0.8057
1.265	23.1899	18.7508	0.8086
1.270	23.1499	18.7203	0.8087
1.275	23.0971	18.7206	0.8105
1.280	23.0169	18.6998	0.8124
1.285	22.9680	18.7351	0.8157
1.290	22.8953	18.6319	0.8138
1.295	22.8102	18.6229	0.8164
1.300	22.7555	18.6191	0.8182
1.305	22.6977	18.5919	0.8191
1.310	22.6281	18.5767	0.8210
1.315	22.5739	18.5648	0.8224
1.320	22.5001	18.4914	0.8218
1.325	22.4269	18.4887	0.8244
1.330	22.3524	18.4567	0.8257
1.335	22.3076	18.4404	0.8266
1.340	22.2779	18.4241	0.8270
1.345	22.1807	18.3741	0.8284
1.350	22.1164	18.3789	0.8310
1.355	22.0795	18.3080	0.8292
1.360	22.0386	18.3496	0.8326
1.365	21.9639	18.3286	0.8345
1.370	21.9255	18.3024	0.8348
1.375	21.8883	18.3115	0.8366
1.380	21.7945	18.2780	0.8387
1.385	21.7358	18.2752	0.8408
1.390	21.6833	18.2341	0.8409
1.395	21.6203	18.2499	0.8441
1.400	21.6030	18.2240	0.8436
1.405	21.4924	18.1791	0.8458
1.410	21.4730	18.1890	0.8471
1.415	21.4109	18.1078	0.8457
1.420	21.3722	18.1349	0.8485
1.425	21.3320	18.0682	0.8470
1.430	21.2269	18.0586	0.8507
1.435	21.2049	18.0170	0.8497
1.440	21.1336	18.0270	0.8530
1.445	21.0892	18.0017	0.8536
1.450	21.0531	18.0128	0.8556
1.455	20.9747	18.0023	0.8583
1.460	20.9130	18.0027	0.8608
1.465	20.8507	17.9615	0.8614

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.470	20.8110	17.9186	0.8610
1.475	20.7925	17.8449	0.8582
1.480	20.7436	17.8857	0.8622
1.485	20.6752	17.8608	0.8639
1.490	20.6510	17.8521	0.8645
1.495	20.5783	17.8149	0.8657
1.500	20.5038	17.8108	0.8687
1.505	20.4500	17.8015	0.8705
1.510	20.3996	17.7330	0.8693
1.515	20.3719	17.8006	0.8738
1.520	20.2924	17.7483	0.8746
1.525	20.2493	17.7410	0.8761
1.530	20.2248	17.6979	0.8751
1.535	20.1834	17.6752	0.8757
1.540	20.1218	17.6529	0.8773
1.545	20.0403	17.6596	0.8812
1.550	20.0486	17.6417	0.8799
1.555	19.9996	17.5975	0.8799
1.560	19.9221	17.5847	0.8827
1.565	19.8616	17.5762	0.8849
1.570	19.8338	17.5321	0.8840
1.575	19.7832	17.5454	0.8869
1.580	19.7739	17.5088	0.8855
1.585	19.7160	17.4665	0.8859
1.590	19.6523	17.4613	0.8885
1.595	19.6182	17.4429	0.8891
1.600	19.5699	17.4368	0.8910
1.605	19.5453	17.4137	0.8909
1.610	19.5063	17.3615	0.8900
1.615	19.4489	17.3742	0.8933
1.620	19.3978	17.3837	0.8962
1.625	19.3501	17.3757	0.8980
1.630	19.3143	17.3042	0.8959
1.635	19.2766	17.3126	0.8981
1.640	19.2493	17.3047	0.8990
1.645	19.1860	17.2672	0.9000
1.650	19.1439	17.2204	0.8995
1.655	19.1217	17.2111	0.9001
1.660	19.0343	17.2042	0.9039
1.665	19.0168	17.2743	0.9084
1.670	18.9690	17.2036	0.9069
1.675	18.9283	17.1646	0.9068
1.680	18.9288	17.1735	0.9073
1.685	18.9066	17.0744	0.9031
1.690	18.8537	17.1130	0.9077
1.695	18.8348	17.0747	0.9066

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.700	18.7863	17.0554	0.9079
1.705	18.7385	17.0497	0.9099
1.710	18.7110	17.0116	0.9092
1.715	18.7109	16.9649	0.9067
1.720	18.6210	16.9644	0.9110
1.725	18.5967	16.9298	0.9104
1.730	18.5972	16.9593	0.9119
1.735	18.5335	16.9191	0.9129
1.740	18.4854	16.9142	0.9150
1.745	18.4099	16.8830	0.9171
1.750	18.4635	16.8513	0.9127
1.755	18.4057	16.8903	0.9177
1.760	18.3725	16.8575	0.9175
1.765	18.3500	16.8616	0.9189
1.770	18.3051	16.8281	0.9193
1.775	18.2618	16.8642	0.9235
1.780	18.2429	16.7908	0.9204
1.785	18.1721	16.7828	0.9235
1.790	18.1739	16.7449	0.9214
1.795	18.1477	16.7371	0.9223
1.800	18.1225	16.6977	0.9214
1.805	18.0567	16.6780	0.9236
1.810	18.0446	16.6621	0.9234
1.815	18.0141	16.6435	0.9239
1.820	17.9813	16.6348	0.9251
1.825	17.9559	16.6229	0.9258
1.830	17.8949	16.5922	0.9272
1.835	17.8639	16.5917	0.9288
1.840	17.8283	16.5719	0.9295
1.845	17.8035	16.5543	0.9298
1.850	17.7713	16.5340	0.9304
1.855	17.7578	16.5159	0.9301
1.860	17.7260	16.4708	0.9292
1.865	17.7089	16.4875	0.9310
1.870	17.6750	16.4737	0.9320
1.875	17.6252	16.4508	0.9334
1.880	17.6048	16.4186	0.9326
1.885	17.5524	16.3805	0.9332
1.890	17.5257	16.3668	0.9339
1.895	17.4864	16.3496	0.9350
1.900	17.4315	16.3694	0.9391
1.905	17.4313	16.3257	0.9366
1.910	17.3920	16.2989	0.9371
1.915	17.3422	16.2918	0.9394
1.920	17.3248	16.2606	0.9386
1.925	17.2884	16.2659	0.9409

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.930	17.2855	16.2069	0.9376
1.935	17.2409	16.2166	0.9406
1.940	17.2149	16.1357	0.9373
1.945	17.1345	16.1656	0.9434
1.950	17.1135	16.1377	0.9430
1.955	17.0793	16.1326	0.9446
1.960	17.0784	16.1044	0.9430
1.965	17.0151	16.0744	0.9447
1.970	17.0021	16.0589	0.9445
1.975	16.9486	16.0765	0.9485
1.980	16.9248	16.0102	0.9460
1.985	16.9259	16.0042	0.9455
1.990	16.8728	15.9893	0.9476
1.995	16.8420	15.9719	0.9483
2.000	16.7765	15.9498	0.9507
2.005	16.7931	15.9360	0.9490
2.010	16.7175	15.9460	0.9538
2.015	16.7042	15.8746	0.9503
2.020	16.6653	15.8622	0.9518
2.025	16.6458	15.8730	0.9536
2.030	16.6531	15.8527	0.9519
2.035	16.5899	15.8505	0.9554
2.040	16.5730	15.8330	0.9553
2.045	16.5080	15.7616	0.9548
2.050	16.4824	15.8286	0.9603
2.055	16.4714	15.8056	0.9596
2.060	16.4130	15.7741	0.9611
2.065	16.4177	15.7737	0.9608
2.070	16.3690	15.7166	0.9601
2.075	16.3315	15.6876	0.9606
2.080	16.3106	15.6885	0.9619
2.085	16.2875	15.6840	0.9629
2.090	16.2575	15.6678	0.9637
2.095	16.2347	15.6485	0.9639
2.100	16.1673	15.6519	0.9681
2.105	16.1559	15.6246	0.9671
2.110	16.1408	15.6299	0.9684
2.115	16.1029	15.5852	0.9679
2.120	16.0744	15.5366	0.9665
2.125	16.0261	15.5380	0.9695
2.130	16.0206	15.5252	0.9691
2.135	15.9907	15.5211	0.9706
2.140	15.9775	15.5041	0.9704
2.145	15.9330	15.4664	0.9707
2.150	15.9346	15.4684	0.9707
2.155	15.8845	15.4431	0.9722

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.160	15.8358	15.4240	0.9740
2.165	15.8337	15.4316	0.9746
2.170	15.7887	15.3965	0.9752
2.175	15.7691	15.4066	0.9770
2.180	15.7368	15.3813	0.9774
2.185	15.7166	15.3745	0.9782
2.190	15.6845	15.3445	0.9783
2.195	15.6197	15.3029	0.9797
2.200	15.6060	15.3008	0.9804
2.205	15.6067	15.2694	0.9784
2.210	15.5721	15.2995	0.9825
2.215	15.5524	15.2530	0.9807
2.220	15.5208	15.2488	0.9825
2.225	15.4767	15.2465	0.9851
2.230	15.4490	15.2285	0.9857
2.235	15.4335	15.2154	0.9859
2.240	15.4288	15.1764	0.9836
2.245	15.3667	15.1753	0.9875
2.250	15.3592	15.1511	0.9865
2.255	15.3555	15.1423	0.9861
2.260	15.2945	15.1283	0.9891
2.265	15.2715	15.1243	0.9904
2.270	15.2714	15.0726	0.9870
2.275	15.2226	15.0510	0.9887
2.280	15.2239	15.0327	0.9874
2.285	15.1933	15.0535	0.9908
2.290	15.1714	15.0098	0.9893
2.295	15.1807	15.0017	0.9882
2.300	15.1617	14.9551	0.9864
2.305	15.0831	14.9710	0.9926
2.310	15.1166	14.9138	0.9866
2.315	15.0902	14.9396	0.9900
2.320	15.0500	14.9252	0.9917
2.325	15.0263	14.8654	0.9893
2.330	14.9975	14.9206	0.9949
2.335	14.9719	14.8786	0.9938
2.340	14.9995	14.8451	0.9897
2.345	14.9192	14.8400	0.9947
2.350	14.9051	14.8353	0.9953
2.355	14.9062	14.7770	0.9913
2.360	14.8728	14.8090	0.9957
2.365	14.8577	14.7965	0.9959
2.370	14.8466	14.7685	0.9947
2.375	14.8426	14.7316	0.9925
2.380	14.8090	14.7535	0.9963
2.385	14.7722	14.7246	0.9968

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.390	14.7660	14.7299	0.9976
2.395	14.7119	14.7145	1.0002
2.400	14.6883	14.6877	1.0000
2.405	14.6957	14.6476	0.9967
2.410	14.6815	14.6559	0.9983
2.415	14.6466	14.6301	0.9989
2.420	14.6200	14.6429	1.0016
2.425	14.6067	14.5913	0.9989
2.430	14.5953	14.5719	0.9984
2.435	14.5990	14.5758	0.9984
2.440	14.5680	14.5543	0.9991
2.445	14.5488	14.5509	1.0001
2.450	14.5375	14.5406	1.0002
2.455	14.5122	14.5074	0.9997
2.460	14.4996	14.5061	1.0005
2.465	14.4796	14.5188	1.0027
2.470	14.4583	14.4655	1.0005
2.475	14.4443	14.4571	1.0009
2.480	14.4181	14.4438	1.0018
2.485	14.4068	14.4537	1.0033
2.490	14.3868	14.4232	1.0025
2.495	14.3554	14.4110	1.0039
2.500	14.3685	14.4051	1.0025
2.505	14.3440	14.3741	1.0021
2.510	14.3207	14.3712	1.0035
2.515	14.2994	14.3485	1.0034
2.520	14.2787	14.3486	1.0049
2.525	14.2415	14.3092	1.0048
2.530	14.2390	14.3043	1.0046
2.535	14.2402	14.3072	1.0047
2.540	14.2186	14.2680	1.0035
2.545	14.2051	14.2454	1.0028
2.550	14.1680	14.2378	1.0049
2.555	14.1722	14.2304	1.0041
2.560	14.1333	14.2186	1.0060
2.565	14.1101	14.1930	1.0059
2.570	14.1022	14.1755	1.0052
2.575	14.0836	14.1721	1.0063
2.580	14.0510	14.1382	1.0062
2.585	14.0190	14.1508	1.0094
2.590	14.0240	14.0974	1.0052
2.595	13.9981	14.1037	1.0075
2.600	13.9657	14.0662	1.0072
2.605	13.9338	14.0352	1.0073
2.610	13.9568	14.0477	1.0065
2.615	13.9425	14.0093	1.0048

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.620	13.8946	14.0052	1.0080
2.625	13.8747	13.9768	1.0074
2.630	13.8633	13.9896	1.0091
2.635	13.8677	13.9609	1.0067
2.640	13.8124	13.9346	1.0088
2.645	13.8169	13.9258	1.0079
2.650	13.7701	13.9148	1.0105
2.655	13.7871	13.8909	1.0075
2.660	13.7486	13.8829	1.0098
2.665	13.7648	13.8855	1.0088
2.670	13.7498	13.8463	1.0070
2.675	13.7170	13.8399	1.0090
2.680	13.7012	13.8348	1.0098
2.685	13.6591	13.8103	1.0111
2.690	13.6462	13.7984	1.0112
2.695	13.6289	13.7978	1.0124
2.700	13.6126	13.7900	1.0130
2.705	13.5975	13.7760	1.0131
2.710	13.5649	13.7833	1.0161
2.715	13.5699	13.7539	1.0136
2.720	13.5295	13.7531	1.0165
2.725	13.4978	13.7142	1.0160
2.730	13.4621	13.6977	1.0175
2.735	13.4485	13.7156	1.0199
2.740	13.4681	13.6622	1.0144
2.745	13.4165	13.6992	1.0211
2.750	13.3946	13.6798	1.0213
2.755	13.3699	13.6583	1.0216
2.760	13.3790	13.6663	1.0215
2.765	13.3549	13.6180	1.0197
2.770	13.3420	13.6584	1.0237
2.775	13.3170	13.5963	1.0210
2.780	13.3147	13.5946	1.0210
2.785	13.2843	13.5921	1.0232
2.790	13.2888	13.5695	1.0211
2.795	13.2532	13.5765	1.0244
2.800	13.2430	13.5508	1.0232
2.805	13.2318	13.5523	1.0242
2.810	13.1955	13.5131	1.0241
2.815	13.1861	13.5194	1.0253
2.820	13.1801	13.5024	1.0245
2.825	13.1630	13.4788	1.0240
2.830	13.1448	13.4822	1.0257
2.835	13.1408	13.4715	1.0252
2.840	13.1036	13.4572	1.0270
2.845	13.0856	13.4503	1.0279

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.850	13.1049	13.4049	1.0229
2.855	13.0702	13.4235	1.0270
2.860	13.0504	13.3887	1.0259
2.865	13.0139	13.3884	1.0288
2.870	13.0046	13.3723	1.0283
2.875	13.0006	13.3526	1.0271
2.880	12.9689	13.3615	1.0303
2.885	12.9746	13.3542	1.0293
2.890	12.9211	13.3589	1.0339
2.895	12.9358	13.3158	1.0294
2.900	12.9371	13.3089	1.0287
2.905	12.8921	13.3253	1.0336
2.910	12.8890	13.3208	1.0335
2.915	12.8827	13.2898	1.0316
2.920	12.8740	13.2868	1.0321
2.925	12.8832	13.2513	1.0286
2.930	12.8656	13.2366	1.0288
2.935	12.8429	13.2242	1.0297
2.940	12.8370	13.2264	1.0303
2.945	12.8082	13.2103	1.0314
2.950	12.7846	13.1836	1.0312
2.955	12.8286	13.1946	1.0285
2.960	12.7079	13.1683	1.0362
2.965	12.7719	13.1393	1.0288
2.970	12.7332	13.1001	1.0288
2.975	12.7236	13.1016	1.0297
2.980	12.7225	13.1174	1.0310
2.985	12.6894	13.0849	1.0312
2.990	12.6555	13.0913	1.0344
2.995	12.6673	13.0440	1.0297
3.000	12.6198	13.0493	1.0340
3.005	12.6417	13.0399	1.0315
3.010	12.6352	12.9860	1.0278
3.015	12.6095	12.9802	1.0294
3.020	12.5639	13.0073	1.0353
3.025	12.5621	12.9720	1.0326
3.030	12.5792	12.9557	1.0299
3.035	12.5627	12.9908	1.0341
3.040	12.5240	12.9614	1.0349
3.045	12.5047	12.9433	1.0351
3.050	12.4785	12.8960	1.0335
3.055	12.4767	12.9701	1.0395
3.060	12.4662	12.9511	1.0389
3.065	12.5033	12.9268	1.0339
3.070	12.4410	12.9418	1.0403
3.075	12.4596	12.9343	1.0381

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.080	12.4385	12.9366	1.0400
3.085	12.4342	12.8858	1.0363
3.090	12.4404	12.8980	1.0368
3.095	12.4330	12.8937	1.0371
3.100	12.4404	12.8812	1.0354
3.105	12.4253	12.8397	1.0334
3.110	12.4196	12.8513	1.0348
3.115	12.4129	12.8320	1.0338
3.120	12.3893	12.8305	1.0356
3.125	12.3808	12.7980	1.0337
3.130	12.3675	12.7822	1.0335
3.135	12.3426	12.7567	1.0336
3.140	12.3533	12.7491	1.0320
3.145	12.3162	12.7133	1.0322
3.150	12.3188	12.7160	1.0322
3.155	12.2917	12.7041	1.0336
3.160	12.2987	12.6894	1.0318
3.165	12.2538	12.6748	1.0344
3.170	12.2846	12.6523	1.0299
3.175	12.2350	12.6290	1.0322
3.180	12.2366	12.6288	1.0320
3.185	12.1993	12.6350	1.0357
3.190	12.2020	12.6199	1.0343
3.195	12.1626	12.6132	1.0371
3.200	12.1293	12.5932	1.0382
3.205	12.1348	12.6167	1.0397
3.210	12.1287	12.5992	1.0388
3.215	12.1440	12.5864	1.0364
3.220	12.1525	12.6084	1.0375
3.225	12.0888	12.5992	1.0422
3.230	12.1245	12.5403	1.0343
3.235	12.0891	12.5590	1.0389
3.240	12.0575	12.5292	1.0391
3.245	12.0765	12.4987	1.0350
3.250	11.9948	12.5332	1.0449
3.255	11.9985	12.4719	1.0394
3.260	12.0255	12.4842	1.0381
3.265	11.9917	12.5040	1.0427
3.270	11.9942	12.4009	1.0339
3.275	11.9595	12.4124	1.0379
3.280	11.9644	12.4185	1.0380
3.285	11.8793	12.3957	1.0435
3.290	11.9381	12.3703	1.0362
3.295	11.8954	12.3444	1.0377
3.300	11.9183	12.3569	1.0368
3.305	11.8769	12.3541	1.0402

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.310	11.8696	12.3255	1.0384
3.315	11.8501	12.3055	1.0384
3.320	11.8470	12.3276	1.0406
3.325	11.8472	12.3092	1.0390
3.330	11.8089	12.3014	1.0417
3.335	11.7768	12.2420	1.0395
3.340	11.7829	12.2624	1.0407
3.345	11.7605	12.2256	1.0395
3.350	11.7244	12.2175	1.0421
3.355	11.7456	12.2141	1.0399
3.360	11.6972	12.2121	1.0440
3.365	11.6865	12.1965	1.0436
3.370	11.6590	12.1905	1.0456
3.375	11.6853	12.2064	1.0446
3.380	11.6669	12.1906	1.0449
3.385	11.6399	12.1749	1.0460
3.390	11.6337	12.1552	1.0448
3.395	11.6298	12.1583	1.0454
3.400	11.5728	12.1478	1.0497
3.405	11.5825	12.1250	1.0468
3.410	11.5743	12.1456	1.0494
3.415	11.5363	12.1143	1.0501
3.420	11.5550	12.1020	1.0473
3.425	11.5696	12.0976	1.0456
3.430	11.5624	12.0982	1.0463
3.435	11.5346	12.1105	1.0499
3.440	11.5482	12.0613	1.0444
3.445	11.5058	12.0645	1.0486
3.450	11.4655	12.0270	1.0490
3.455	11.4897	12.0517	1.0489
3.460	11.4704	12.0272	1.0485
3.465	11.4698	11.9977	1.0460
3.470	11.4586	11.9896	1.0463
3.475	11.4757	11.9584	1.0421
3.480	11.4599	11.9183	1.0400
3.485	11.4825	11.8877	1.0353
3.490	11.4020	11.9374	1.0470
3.495	11.3929	11.9224	1.0465
3.500	11.3974	11.8730	1.0417
3.505	11.4090	11.9049	1.0435
3.510	11.3977	11.8247	1.0375
3.515	11.3523	11.8545	1.0442
3.520	11.3330	11.8150	1.0425
3.525	11.2217	11.8179	1.0531
3.530	11.2564	11.8493	1.0527
3.535	11.2584	11.7964	1.0478

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.540	11.1943	11.8100	1.0550
3.545	11.2202	11.8760	1.0585
3.550	11.1749	11.8592	1.0612
3.555	11.1989	11.8442	1.0576
3.560	11.2025	11.8807	1.0605
3.565	11.2336	11.8793	1.0575
3.570	11.1669	11.8821	1.0640
3.575	11.1616	11.9587	1.0714
3.580	11.2322	11.9478	1.0637
3.585	11.3433	11.9288	1.0516
3.590	11.3597	11.8092	1.0396
3.595	11.3841	11.7789	1.0347
3.600	11.3310	11.7916	1.0406
3.605	11.3787	11.7056	1.0287
3.610	11.3443	11.6700	1.0287
3.615	11.2775	11.6444	1.0325
3.620	11.2300	11.6024	1.0332
3.625	11.2928	11.5576	1.0234
3.630	11.1401	11.5291	1.0349
3.635	11.1328	11.5656	1.0389
3.640	11.1411	11.5917	1.0404
3.645	11.1071	11.5941	1.0438
3.650	11.0598	11.5896	1.0479
3.655	11.0533	11.5683	1.0466
3.660	11.0336	11.6072	1.0520
3.665	11.0378	11.5752	1.0487
3.670	11.0024	11.5920	1.0536
3.675	11.0148	11.5607	1.0496
3.680	11.0100	11.5858	1.0523
3.685	10.9998	11.6043	1.0550
3.690	10.9919	11.5998	1.0553
3.695	10.9943	11.6193	1.0569
3.700	11.0133	11.6144	1.0546
3.705	11.0343	11.6120	1.0523
3.710	11.0099	11.6016	1.0537
3.715	11.0531	11.6013	1.0496
3.720	11.0990	11.5710	1.0425
3.725	11.0825	11.5680	1.0438
3.730	11.1088	11.5272	1.0377
3.735	11.1084	11.4582	1.0315
3.740	11.1157	11.4222	1.0276
3.745	11.0959	11.3903	1.0265
3.750	11.0293	11.3318	1.0274
3.755	10.9727	11.3099	1.0307
3.760	10.8975	11.3069	1.0376
3.765	10.8135	11.3293	1.0477

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.770	10.8094	11.4084	1.0554
3.775	10.8339	11.4155	1.0537
3.780	10.6509	11.5264	1.0822
3.785	10.8407	11.5264	1.0633
3.790	10.9212	11.4689	1.0501
3.795	10.9263	11.4474	1.0477
3.800	10.9058	11.4823	1.0529
3.805	10.9571	11.4241	1.0426
3.810	10.9181	11.3825	1.0425
3.815	10.9122	11.3118	1.0366
3.820	10.9178	11.3465	1.0393
3.825	10.8899	11.2845	1.0362
3.830	10.8854	11.3018	1.0382
3.835	10.6792	10.8645	1.0174
3.840	10.8290	11.2700	1.0407
3.845	10.8117	11.2691	1.0423
3.850	10.8070	11.2512	1.0411
3.855	10.7859	11.2544	1.0434
3.860	10.7821	11.2231	1.0409
3.865	10.7340	11.2188	1.0452
3.870	10.7741	11.2233	1.0417
3.875	10.7377	11.2063	1.0436
3.880	10.7558	11.2000	1.0413
3.885	10.7155	11.1937	1.0446
3.890	10.7241	11.2090	1.0452
3.895	10.7331	11.1638	1.0401
3.900	10.7061	11.1538	1.0418
3.905	10.6695	11.1590	1.0459
3.910	10.6922	11.1452	1.0424
3.915	10.6747	11.1367	1.0433
3.920	10.6530	11.1445	1.0461
3.925	10.6512	11.1079	1.0429
3.930	10.6461	11.1013	1.0428
3.935	10.6492	11.1014	1.0425
3.940	10.6199	11.0874	1.0440
3.945	10.6213	11.0679	1.0421
3.950	10.6008	11.0533	1.0427
3.955	10.5828	11.0609	1.0452
3.960	10.5844	11.0379	1.0428
3.965	10.5796	11.0367	1.0432
3.970	10.5578	11.0361	1.0453
3.975	10.5480	11.0193	1.0447
3.980	10.5480	11.0052	1.0433
3.985	10.5431	11.0121	1.0445
3.990	10.5400	10.9999	1.0436
3.995	10.5236	10.9781	1.0432

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.000	10.5096	10.9710	1.0439
4.005	10.4957	10.9606	1.0443
4.010	10.4802	10.9553	1.0453
4.015	10.4783	10.9591	1.0459
4.020	10.4710	10.9377	1.0446
4.025	10.4897	10.9280	1.0418
4.030	10.4466	10.9125	1.0446
4.035	10.4296	10.9186	1.0469
4.040	10.4434	10.9074	1.0444
4.045	10.4404	10.8840	1.0425
4.050	10.4207	10.8688	1.0430
4.055	10.4084	10.8574	1.0431
4.060	10.3928	10.8547	1.0444
4.065	10.3686	10.8432	1.0458
4.070	10.3803	10.8565	1.0459
4.075	10.3724	10.8351	1.0446
4.080	10.3513	10.8535	1.0485
4.085	10.3516	10.8246	1.0457
4.090	10.3539	10.8133	1.0444
4.095	10.3261	10.8107	1.0469
4.100	10.3284	10.8258	1.0482
4.105	10.3269	10.7523	1.0412
4.110	10.3208	10.7906	1.0455
4.115	10.3041	10.7768	1.0459
4.120	10.2883	10.7662	1.0465
4.125	10.3110	10.7740	1.0449
4.130	10.2994	10.7815	1.0468
4.135	10.2646	10.7378	1.0461
4.140	10.2889	10.7078	1.0407
4.145	10.3179	10.7121	1.0382
4.150	10.2373	10.7066	1.0458
4.155	10.2238	10.7012	1.0467
4.160	10.2506	10.6998	1.0438
4.165	10.2268	10.6533	1.0417
4.170	10.2215	10.6483	1.0418
4.175	10.2413	10.6275	1.0377
4.180	10.2369	10.6513	1.0405
4.185	10.2345	10.6046	1.0362
4.190	10.2080	10.6029	1.0387
4.195	10.2066	10.6005	1.0386
4.200	10.2235	10.6047	1.0373
4.205	10.2176	10.6034	1.0378
4.210	10.1698	10.5897	1.0413
4.215	10.1980	10.6022	1.0396
4.220	10.1838	10.5948	1.0404
4.225	10.2275	10.6037	1.0368

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.230	10.1897	10.5839	1.0387
4.235	10.1702	10.5755	1.0399
4.240	10.1584	10.6148	1.0449
4.245	10.1825	10.5854	1.0396
4.250	10.1711	10.5647	1.0387
4.255	10.1704	10.5883	1.0411
4.260	10.1417	10.5637	1.0416
4.265	10.1777	10.5627	1.0378
4.270	10.1414	10.5527	1.0406
4.275	10.1332	10.5465	1.0408
4.280	10.1284	10.5520	1.0418
4.285	10.1149	10.5490	1.0429
4.290	10.1079	10.5359	1.0423
4.295	10.1068	10.5269	1.0416
4.300	10.1019	10.5391	1.0433
4.305	10.0943	10.5280	1.0430
4.310	10.0938	10.5135	1.0416
4.315	10.0883	10.5134	1.0421
4.320	10.0914	10.5043	1.0409
4.325	10.0829	10.5018	1.0415
4.330	10.0876	10.4998	1.0409
4.335	10.0804	10.4952	1.0411
4.340	10.0861	10.4789	1.0390
4.345	10.0687	10.4658	1.0394
4.350	10.0713	10.4622	1.0388
4.355	10.0587	10.4618	1.0401
4.360	10.0502	10.4413	1.0389
4.365	10.0251	10.4303	1.0404
4.370	10.0457	10.4277	1.0380
4.375	10.0071	10.4271	1.0420
4.380	10.0165	10.4163	1.0399
4.385	9.9752	10.3974	1.0423
4.390	9.9993	10.3750	1.0376
4.395	9.9718	10.3750	1.0404
4.400	9.9593	10.3775	1.0420
4.405	9.9276	10.3674	1.0443
4.410	9.9433	10.3729	1.0432
4.415	9.9352	10.3431	1.0411
4.420	9.9190	10.3656	1.0450
4.425	9.9362	10.3693	1.0436
4.430	9.9144	10.3302	1.0419
4.435	9.8926	10.3061	1.0418
4.440	9.8973	10.3212	1.0428
4.445	9.8664	10.2887	1.0428
4.450	9.8834	10.2710	1.0392
4.455	9.8438	10.2800	1.0443

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.460	9.8628	10.2372	1.0380
4.465	9.8729	10.2015	1.0333
4.470	9.8799	10.2199	1.0344
4.475	9.8924	10.1979	1.0309
4.480	9.9086	10.2091	1.0303
4.485	9.8606	10.1635	1.0307
4.490	9.8702	10.1849	1.0319
4.495	9.8722	10.1761	1.0308
4.500	9.8511	10.1692	1.0323
4.505	9.8305	10.1533	1.0328
4.510	9.7911	10.1433	1.0360
4.515	9.7923	10.1787	1.0395
4.520	9.7850	10.1695	1.0393
4.525	9.8059	10.1618	1.0363
4.530	9.7682	10.1382	1.0379
4.535	9.7677	10.1527	1.0394
4.540	9.7728	10.1290	1.0364
4.545	9.7519	10.1178	1.0375
4.550	9.7568	10.0937	1.0345
4.555	9.7323	10.1002	1.0378
4.560	9.7349	10.0999	1.0375
4.565	9.7197	10.1026	1.0394
4.570	9.7172	10.0808	1.0374
4.575	9.7167	10.0753	1.0369
4.580	9.7126	10.0829	1.0381
4.585	9.6996	10.0682	1.0380
4.590	9.7007	10.0524	1.0363
4.595	9.6811	10.0569	1.0388
4.600	9.6793	10.0440	1.0377
4.605	9.6625	10.0286	1.0379
4.610	9.6647	10.0298	1.0378
4.615	9.6509	10.0108	1.0373
4.620	9.6599	9.9965	1.0348
4.625	9.6491	10.0050	1.0369
4.630	9.6413	9.9868	1.0358
4.635	9.6327	9.9722	1.0352
4.640	9.6176	9.9689	1.0365
4.645	9.6240	9.9574	1.0346
4.650	9.6168	9.9571	1.0354
4.655	9.6089	9.9242	1.0328
4.660	9.6195	9.9460	1.0339
4.665	9.6103	9.9463	1.0350
4.670	9.5610	9.9135	1.0369
4.675	9.5746	9.8814	1.0320
4.680	9.5663	9.8889	1.0337
4.685	9.5729	9.8809	1.0322

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.690	9.5657	9.9090	1.0359
4.695	9.5746	9.8622	1.0300
4.700	9.5441	9.8806	1.0353
4.705	9.5352	9.8099	1.0288
4.710	9.4773	9.8548	1.0398
4.715	9.5080	9.8531	1.0363
4.720	9.5010	9.8209	1.0337
4.725	9.4957	9.8562	1.0380
4.730	9.5105	9.8377	1.0344
4.735	9.5136	9.8538	1.0358
4.740	9.5144	9.8413	1.0344
4.745	9.4796	9.8331	1.0373
4.750	9.5164	9.8206	1.0320
4.755	9.5038	9.8421	1.0356
4.760	9.4915	9.8059	1.0331
4.765	9.5266	9.8240	1.0312
4.770	9.5126	9.8347	1.0339
4.775	9.4640	9.7594	1.0312
4.780	9.4873	9.7672	1.0295
4.785	9.4658	9.8470	1.0403
4.790	9.4932	9.7468	1.0267
4.795	9.4517	9.7422	1.0307
4.800	9.4569	9.7466	1.0306
4.805	9.4845	9.7069	1.0235
4.810	9.4526	9.7548	1.0320
4.815	9.4806	9.7025	1.0234
4.820	9.4148	9.7239	1.0328
4.825	9.4434	9.7359	1.0310
4.830	9.4079	9.7207	1.0332
4.835	9.4098	9.7298	1.0340
4.840	9.4173	9.7232	1.0325
4.845	9.4304	9.7069	1.0293
4.850	9.3905	9.7100	1.0340
4.855	9.4230	9.7198	1.0315
4.860	9.4126	9.6989	1.0304
4.865	9.4405	9.6969	1.0272
4.870	9.4101	9.6865	1.0294
4.875	9.4141	9.6956	1.0299
4.880	9.4106	9.7037	1.0311
4.885	9.4116	9.6719	1.0277
4.890	9.4341	9.7025	1.0284
4.895	9.4036	9.6720	1.0285
4.900	9.4084	9.6891	1.0298
4.905	9.4089	9.6760	1.0284
4.910	9.3910	9.6665	1.0293
4.915	9.3882	9.6674	1.0297

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.920	9.3863	9.6606	1.0292
4.925	9.3830	9.6569	1.0292
4.930	9.3630	9.6437	1.0300
4.935	9.3632	9.6430	1.0299
4.940	9.3594	9.6418	1.0302
4.945	9.3419	9.6403	1.0320
4.950	9.3420	9.6295	1.0308
4.955	9.3372	9.6321	1.0316
4.960	9.3259	9.6131	1.0308
4.965	9.3227	9.6077	1.0306
4.970	9.3074	9.6074	1.0322
4.975	9.3139	9.6014	1.0309
4.980	9.2833	9.5912	1.0332
4.985	9.2797	9.5751	1.0318
4.990	9.2693	9.5778	1.0333
4.995	9.2315	9.5639	1.0360
5.000	9.2353	9.5449	1.0335
5.005	9.2189	9.5356	1.0344
5.010	9.2057	9.5114	1.0332
5.015	9.1897	9.5138	1.0353
5.020	9.1902	9.4786	1.0314
5.025	9.1556	9.4553	1.0327
5.030	9.1430	9.4280	1.0312
5.035	9.1686	9.4274	1.0282
5.040	9.1583	9.4012	1.0265
5.045	9.1806	9.3784	1.0215
5.050	9.1786	9.3370	1.0173
5.055	9.2120	9.3514	1.0151
5.060	9.2064	9.3416	1.0147
5.065	9.2263	9.3291	1.0111
5.070	9.2192	9.3556	1.0148
5.075	9.2211	9.3537	1.0144
5.080	9.2121	9.3426	1.0142
5.085	9.2104	9.3382	1.0139
5.090	9.2120	9.3624	1.0163
5.095	9.2191	9.3812	1.0176
5.100	9.2067	9.3771	1.0185
5.105	9.1874	9.4131	1.0246
5.110	9.1760	9.3642	1.0205
5.115	9.1593	9.3583	1.0217
5.120	9.1483	9.3560	1.0227
5.125	9.1416	9.3572	1.0236
5.130	9.1348	9.3387	1.0223
5.135	9.1088	9.3257	1.0238
5.140	9.1057	9.3378	1.0255
5.145	9.0911	9.2974	1.0227

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.150	9.0861	9.3177	1.0255
5.155	9.0771	9.2957	1.0241
5.160	9.0703	9.2869	1.0239
5.165	9.0567	9.2768	1.0243
5.170	9.0546	9.2597	1.0227
5.175	9.0441	9.2441	1.0221
5.180	9.0462	9.2409	1.0215
5.185	9.0364	9.2321	1.0217
5.190	9.0389	9.2237	1.0204
5.195	9.0381	9.2160	1.0197
5.200	9.0351	9.2181	1.0203
5.205	9.0384	9.2138	1.0194
5.210	9.0341	9.2040	1.0188
5.215	9.0328	9.2026	1.0188
5.220	9.0227	9.1896	1.0185
5.225	9.0099	9.1832	1.0192
5.230	9.0215	9.1825	1.0178
5.235	9.0010	9.1839	1.0203
5.240	8.9942	9.1676	1.0193
5.245	9.0020	9.1753	1.0192
5.250	8.9845	9.1761	1.0213
5.255	8.9637	9.1704	1.0231
5.260	9.0004	9.1566	1.0174
5.265	8.9945	9.1764	1.0202
5.270	8.9746	9.1512	1.0197
5.275	8.9652	9.1324	1.0186
5.280	8.9663	9.1464	1.0201
5.285	8.9783	9.1344	1.0174
5.290	8.9583	9.1223	1.0183
5.295	8.9860	9.0917	1.0118
5.300	8.9529	9.1122	1.0178
5.305	8.9737	9.1125	1.0155
5.310	8.9198	9.0908	1.0192
5.315	8.9449	9.0981	1.0171
5.320	8.9284	9.0643	1.0152
5.325	8.9362	9.0825	1.0164
5.330	8.9303	9.0736	1.0161
5.335	8.8990	9.0691	1.0191
5.340	8.8916	9.0840	1.0216
5.345	8.8805	9.0957	1.0242
5.350	8.9241	9.0555	1.0147
5.355	8.8792	9.0574	1.0201
5.360	8.8827	9.0137	1.0147
5.365	8.9231	9.0428	1.0134
5.370	8.8712	9.0064	1.0152
5.375	8.8787	9.0303	1.0171

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.380	8.8827	9.0381	1.0175
5.385	8.8581	9.0104	1.0172
5.390	8.8559	9.0358	1.0203
5.395	8.8825	9.0332	1.0170
5.400	8.8565	9.0158	1.0180
5.405	8.8653	9.0109	1.0164
5.410	8.8488	9.0116	1.0184
5.415	8.8597	8.9851	1.0142
5.420	8.8320	9.0036	1.0194
5.425	8.8208	9.0048	1.0209
5.430	8.8491	8.9858	1.0154
5.435	8.8405	8.9737	1.0151
5.440	8.8089	8.9875	1.0203
5.445	8.8441	8.9825	1.0157
5.450	8.8129	8.9628	1.0170
5.455	8.8286	8.9706	1.0161
5.460	8.8201	8.9619	1.0161
5.465	8.8244	8.9814	1.0178
5.470	8.7869	8.9522	1.0188
5.475	8.8094	8.9473	1.0157
5.480	8.7925	8.9554	1.0185
5.485	8.8009	8.9351	1.0153
5.490	8.7676	8.9503	1.0208
5.495	8.7979	8.9331	1.0154
5.500	8.7853	8.9438	1.0180
5.505	8.7866	8.9446	1.0180
5.510	8.7864	8.9328	1.0167
5.515	8.7906	8.9233	1.0151
5.520	8.7766	8.9145	1.0157
5.525	8.7877	8.9152	1.0145
5.530	8.7791	8.9094	1.0148
5.535	8.7892	8.9018	1.0128
5.540	8.7719	8.9053	1.0152
5.545	8.7628	8.8859	1.0140
5.550	8.7681	8.8853	1.0134
5.555	8.7720	8.8781	1.0121
5.560	8.7645	8.8761	1.0127
5.565	8.7700	8.8721	1.0116
5.570	8.7659	8.8724	1.0121
5.575	8.7730	8.8676	1.0108
5.580	8.7671	8.8789	1.0127
5.585	8.7598	8.8659	1.0121
5.590	8.7728	8.8585	1.0098
5.595	8.7685	8.8531	1.0096
5.600	8.7737	8.8560	1.0094
5.605	8.7860	8.8393	1.0061

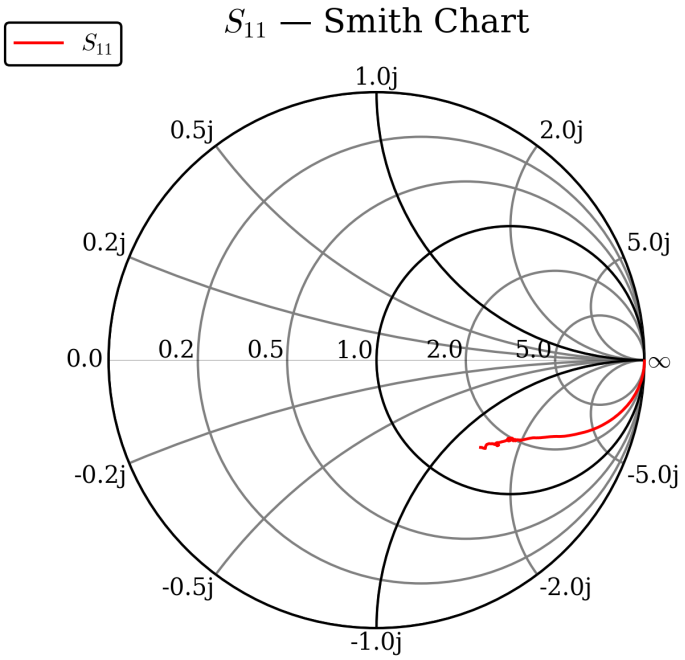
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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.610	8.7707	8.8248	1.0062
5.615	8.7701	8.7995	1.0034
5.620	8.7665	8.8065	1.0046
5.625	8.7630	8.8109	1.0055
5.630	8.7558	8.7859	1.0034
5.635	8.7601	8.7790	1.0022
5.640	8.7616	8.7576	0.9995
5.645	8.7434	8.7452	1.0002
5.650	8.7424	8.7664	1.0027
5.655	8.7291	8.7483	1.0022
5.660	8.7204	8.7500	1.0034
5.665	8.7580	8.7447	0.9985
5.670	8.6893	8.7370	1.0055
5.675	8.7133	8.7097	0.9996
5.680	8.7033	8.7090	1.0007
5.685	8.7241	8.6977	0.9970
5.690	8.7062	8.6928	0.9985
5.695	8.6910	8.7107	1.0023
5.700	8.6898	8.6664	0.9973
5.705	8.6560	8.6631	1.0008
5.710	8.6952	8.6571	0.9956
5.715	8.6717	8.6305	0.9952
5.720	8.6648	8.6410	0.9973
5.725	8.6411	8.6348	0.9993
5.730	8.6477	8.6200	0.9968
5.735	8.6425	8.6187	0.9973
5.740	8.6329	8.6235	0.9989
5.745	8.6356	8.6340	0.9998
5.750	8.6154	8.6057	0.9989
5.755	8.6241	8.6046	0.9977
5.760	8.6012	8.5813	0.9977
5.765	8.6264	8.5805	0.9947
5.770	8.6165	8.5606	0.9935
5.775	8.6134	8.5732	0.9953
5.780	8.5850	8.5712	0.9984
5.785	8.5788	8.5591	0.9977
5.790	8.5641	8.5652	1.0001
5.795	8.5710	8.5667	0.9995
5.800	8.5565	8.5530	0.9996
5.805	8.5547	8.5410	0.9984
5.810	8.5491	8.5506	1.0002
5.815	8.5479	8.5468	0.9999
5.820	8.5451	8.5352	0.9988
5.825	8.5382	8.5307	0.9991
5.830	8.5238	8.5249	1.0001
5.835	8.5301	8.5194	0.9988

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.840	8.5205	8.5169	0.9996
5.845	8.5096	8.5167	1.0008
5.850	8.5134	8.5075	0.9993
5.855	8.5090	8.4957	0.9984
5.860	8.4958	8.4952	0.9999
5.865	8.5068	8.4981	0.9990
5.870	8.4946	8.4833	0.9987
5.875	8.4923	8.4812	0.9987
5.880	8.4872	8.4760	0.9987
5.885	8.4993	8.4581	0.9952
5.890	8.4574	8.4757	1.0022
5.895	8.4651	8.4581	0.9992
5.900	8.4706	8.4400	0.9964
5.905	8.4614	8.4568	0.9994
5.910	8.4629	8.4247	0.9955
5.915	8.4517	8.4317	0.9976
5.920	8.4483	8.4206	0.9967
5.925	8.4475	8.4180	0.9965
5.930	8.4115	8.4199	1.0010
5.935	8.4140	8.4085	0.9993
5.940	8.4137	8.4073	0.9992
5.945	8.4474	8.3878	0.9929
5.950	8.4072	8.4056	0.9998
5.955	8.4123	8.4056	0.9992
5.960	8.4014	8.3919	0.9989
5.965	8.3997	8.3931	0.9992
5.970	8.3925	8.4009	1.0010
5.975	8.4174	8.3475	0.9917
5.980	8.4500	8.3887	0.9927
5.985	8.3797	8.3762	0.9996
5.990	8.3819	8.3579	0.9971
5.995	8.3870	8.3442	0.9949
6.000	8.4064	8.3470	0.9929

1.2 S_{11} — Smith Chart



Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.005	1.0000	-0.0030	0.00
0.010	0.9997	-0.0061	-0.00
0.015	0.9997	-0.0094	-0.00
0.020	0.9996	-0.0126	-0.00
0.025	0.9994	-0.0157	-0.00
0.030	0.9993	-0.0190	-0.00
0.035	0.9989	-0.0220	-0.01
0.040	0.9986	-0.0254	-0.01
0.045	0.9983	-0.0285	-0.01
0.050	0.9979	-0.0315	-0.01
0.055	0.9976	-0.0346	-0.02
0.060	0.9970	-0.0378	-0.02
0.065	0.9966	-0.0410	-0.02
0.070	0.9959	-0.0441	-0.03
0.075	0.9955	-0.0471	-0.03
0.080	0.9948	-0.0502	-0.03
0.085	0.9942	-0.0532	-0.04
0.090	0.9935	-0.0563	-0.04
0.095	0.9928	-0.0594	-0.05
0.100	0.9921	-0.0624	-0.05
0.105	0.9912	-0.0652	-0.06
0.110	0.9903	-0.0683	-0.06
0.115	0.9896	-0.0713	-0.07

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.120	0.9886	-0.0743	-0.08
0.125	0.9878	-0.0773	-0.08
0.130	0.9868	-0.0801	-0.09
0.135	0.9858	-0.0830	-0.09
0.140	0.9847	-0.0858	-0.10
0.145	0.9838	-0.0887	-0.11
0.150	0.9827	-0.0915	-0.11
0.155	0.9816	-0.0942	-0.12
0.160	0.9805	-0.0969	-0.13
0.165	0.9794	-0.0998	-0.14
0.170	0.9783	-0.1025	-0.14
0.175	0.9770	-0.1051	-0.15
0.180	0.9757	-0.1078	-0.16
0.185	0.9745	-0.1105	-0.17
0.190	0.9731	-0.1133	-0.18
0.195	0.9719	-0.1159	-0.19
0.200	0.9706	-0.1184	-0.20
0.205	0.9692	-0.1209	-0.20
0.210	0.9677	-0.1235	-0.22
0.215	0.9663	-0.1259	-0.22
0.220	0.9650	-0.1285	-0.23
0.225	0.9634	-0.1310	-0.24
0.230	0.9620	-0.1334	-0.25
0.235	0.9605	-0.1360	-0.26
0.240	0.9591	-0.1383	-0.27
0.245	0.9573	-0.1407	-0.29
0.250	0.9559	-0.1429	-0.30
0.255	0.9543	-0.1453	-0.31
0.260	0.9528	-0.1476	-0.32
0.265	0.9512	-0.1499	-0.33
0.270	0.9495	-0.1521	-0.34
0.275	0.9480	-0.1543	-0.35
0.280	0.9463	-0.1565	-0.36
0.285	0.9448	-0.1586	-0.37
0.290	0.9430	-0.1607	-0.39
0.295	0.9413	-0.1628	-0.40
0.300	0.9397	-0.1648	-0.41
0.305	0.9380	-0.1669	-0.42
0.310	0.9364	-0.1689	-0.43
0.315	0.9345	-0.1709	-0.45
0.320	0.9329	-0.1728	-0.46
0.325	0.9311	-0.1748	-0.47
0.330	0.9293	-0.1766	-0.48
0.335	0.9276	-0.1786	-0.49
0.340	0.9259	-0.1803	-0.51
0.345	0.9243	-0.1822	-0.52

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.350	0.9224	-0.1840	-0.53
0.355	0.9207	-0.1857	-0.54
0.360	0.9189	-0.1874	-0.56
0.365	0.9171	-0.1892	-0.57
0.370	0.9154	-0.1908	-0.58
0.375	0.9136	-0.1925	-0.60
0.380	0.9117	-0.1942	-0.61
0.385	0.9099	-0.1958	-0.62
0.390	0.9082	-0.1974	-0.64
0.395	0.9066	-0.1989	-0.65
0.400	0.9047	-0.2005	-0.66
0.405	0.9030	-0.2019	-0.67
0.410	0.9013	-0.2035	-0.69
0.415	0.8995	-0.2048	-0.70
0.420	0.8978	-0.2063	-0.71
0.425	0.8961	-0.2077	-0.73
0.430	0.8944	-0.2090	-0.74
0.435	0.8927	-0.2104	-0.75
0.440	0.8909	-0.2117	-0.76
0.445	0.8892	-0.2131	-0.78
0.450	0.8875	-0.2143	-0.79
0.455	0.8858	-0.2156	-0.80
0.460	0.8842	-0.2169	-0.81
0.465	0.8825	-0.2182	-0.83
0.470	0.8807	-0.2194	-0.84
0.475	0.8790	-0.2205	-0.86
0.480	0.8774	-0.2217	-0.87
0.485	0.8758	-0.2228	-0.88
0.490	0.8741	-0.2238	-0.89
0.495	0.8726	-0.2250	-0.90
0.500	0.8709	-0.2260	-0.92
0.505	0.8692	-0.2272	-0.93
0.510	0.8675	-0.2283	-0.94
0.515	0.8660	-0.2293	-0.96
0.520	0.8645	-0.2303	-0.97
0.525	0.8628	-0.2313	-0.98
0.530	0.8612	-0.2322	-0.99
0.535	0.8597	-0.2331	-1.01
0.540	0.8581	-0.2341	-1.02
0.545	0.8566	-0.2351	-1.03
0.550	0.8550	-0.2360	-1.04
0.555	0.8535	-0.2369	-1.05
0.560	0.8518	-0.2377	-1.07
0.565	0.8504	-0.2386	-1.08
0.570	0.8488	-0.2394	-1.09
0.575	0.8474	-0.2404	-1.10

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.580	0.8458	-0.2412	-1.12
0.585	0.8443	-0.2420	-1.13
0.590	0.8427	-0.2427	-1.14
0.595	0.8413	-0.2435	-1.15
0.600	0.8399	-0.2443	-1.16
0.605	0.8385	-0.2451	-1.17
0.610	0.8369	-0.2457	-1.19
0.615	0.8355	-0.2464	-1.20
0.620	0.8340	-0.2473	-1.21
0.625	0.8327	-0.2480	-1.22
0.630	0.8312	-0.2486	-1.23
0.635	0.8298	-0.2493	-1.25
0.640	0.8284	-0.2500	-1.26
0.645	0.8270	-0.2507	-1.27
0.650	0.8256	-0.2513	-1.28
0.655	0.8242	-0.2520	-1.29
0.660	0.8229	-0.2524	-1.30
0.665	0.8215	-0.2532	-1.31
0.670	0.8200	-0.2538	-1.33
0.675	0.8188	-0.2543	-1.34
0.680	0.8173	-0.2550	-1.35
0.685	0.8159	-0.2555	-1.36
0.690	0.8147	-0.2560	-1.37
0.695	0.8133	-0.2566	-1.38
0.700	0.8118	-0.2571	-1.40
0.705	0.8105	-0.2575	-1.41
0.710	0.8093	-0.2581	-1.42
0.715	0.8079	-0.2587	-1.43
0.720	0.8065	-0.2591	-1.44
0.725	0.8052	-0.2597	-1.45
0.730	0.8039	-0.2601	-1.46
0.735	0.8025	-0.2606	-1.48
0.740	0.8010	-0.2611	-1.49
0.745	0.7998	-0.2616	-1.50
0.750	0.7986	-0.2620	-1.51
0.755	0.7973	-0.2624	-1.52
0.760	0.7960	-0.2629	-1.53
0.765	0.7948	-0.2633	-1.54
0.770	0.7933	-0.2637	-1.56
0.775	0.7920	-0.2642	-1.57
0.780	0.7907	-0.2645	-1.58
0.785	0.7895	-0.2650	-1.59
0.790	0.7881	-0.2656	-1.60
0.795	0.7868	-0.2659	-1.61
0.800	0.7855	-0.2662	-1.62
0.805	0.7841	-0.2666	-1.64

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.810	0.7829	-0.2670	-1.65
0.815	0.7816	-0.2673	-1.66
0.820	0.7803	-0.2678	-1.67
0.825	0.7791	-0.2681	-1.68
0.830	0.7778	-0.2685	-1.69
0.835	0.7764	-0.2689	-1.71
0.840	0.7753	-0.2692	-1.72
0.845	0.7740	-0.2695	-1.73
0.850	0.7727	-0.2699	-1.74
0.855	0.7714	-0.2702	-1.75
0.860	0.7702	-0.2706	-1.76
0.865	0.7689	-0.2709	-1.77
0.870	0.7676	-0.2712	-1.79
0.875	0.7663	-0.2715	-1.80
0.880	0.7650	-0.2719	-1.81
0.885	0.7638	-0.2721	-1.82
0.890	0.7625	-0.2725	-1.83
0.895	0.7614	-0.2727	-1.84
0.900	0.7600	-0.2730	-1.86
0.905	0.7587	-0.2733	-1.87
0.910	0.7574	-0.2736	-1.88
0.915	0.7563	-0.2739	-1.89
0.920	0.7550	-0.2743	-1.90
0.925	0.7537	-0.2745	-1.92
0.930	0.7525	-0.2748	-1.93
0.935	0.7513	-0.2751	-1.94
0.940	0.7500	-0.2754	-1.95
0.945	0.7489	-0.2756	-1.96
0.950	0.7476	-0.2759	-1.97
0.955	0.7464	-0.2762	-1.98
0.960	0.7451	-0.2765	-2.00
0.965	0.7439	-0.2766	-2.01
0.970	0.7428	-0.2770	-2.02
0.975	0.7416	-0.2773	-2.03
0.980	0.7404	-0.2775	-2.04
0.985	0.7392	-0.2777	-2.05
0.990	0.7379	-0.2780	-2.06
0.995	0.7368	-0.2783	-2.07
1.000	0.7356	-0.2784	-2.09
1.005	0.7344	-0.2786	-2.10
1.010	0.7332	-0.2788	-2.11
1.015	0.7321	-0.2791	-2.12
1.020	0.7309	-0.2793	-2.13
1.025	0.7298	-0.2794	-2.14
1.030	0.7287	-0.2797	-2.15
1.035	0.7275	-0.2800	-2.16

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.040	0.7265	-0.2801	-2.17
1.045	0.7253	-0.2803	-2.19
1.050	0.7243	-0.2805	-2.20
1.055	0.7232	-0.2806	-2.21
1.060	0.7222	-0.2808	-2.22
1.065	0.7211	-0.2810	-2.23
1.070	0.7200	-0.2813	-2.24
1.075	0.7191	-0.2814	-2.25
1.080	0.7180	-0.2815	-2.26
1.085	0.7170	-0.2817	-2.27
1.090	0.7160	-0.2818	-2.28
1.095	0.7150	-0.2820	-2.29
1.100	0.7141	-0.2821	-2.30
1.105	0.7129	-0.2823	-2.31
1.110	0.7120	-0.2825	-2.32
1.115	0.7111	-0.2827	-2.32
1.120	0.7100	-0.2827	-2.34
1.125	0.7091	-0.2828	-2.35
1.130	0.7082	-0.2830	-2.35
1.135	0.7073	-0.2832	-2.36
1.140	0.7064	-0.2833	-2.37
1.145	0.7055	-0.2834	-2.38
1.150	0.7047	-0.2835	-2.39
1.155	0.7037	-0.2836	-2.40
1.160	0.7028	-0.2837	-2.41
1.165	0.7021	-0.2839	-2.41
1.170	0.7011	-0.2840	-2.42
1.175	0.7005	-0.2841	-2.43
1.180	0.6996	-0.2842	-2.44
1.185	0.6988	-0.2842	-2.45
1.190	0.6981	-0.2843	-2.46
1.195	0.6973	-0.2843	-2.46
1.200	0.6965	-0.2844	-2.47
1.205	0.6957	-0.2845	-2.48
1.210	0.6950	-0.2845	-2.49
1.215	0.6942	-0.2845	-2.50
1.220	0.6936	-0.2845	-2.50
1.225	0.6928	-0.2847	-2.51
1.230	0.6921	-0.2846	-2.52
1.235	0.6914	-0.2847	-2.53
1.240	0.6906	-0.2847	-2.53
1.245	0.6900	-0.2848	-2.54
1.250	0.6893	-0.2847	-2.55
1.255	0.6886	-0.2847	-2.55
1.260	0.6879	-0.2847	-2.56
1.265	0.6873	-0.2848	-2.57

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.270	0.6867	-0.2848	-2.58
1.275	0.6860	-0.2848	-2.58
1.280	0.6853	-0.2848	-2.59
1.285	0.6847	-0.2849	-2.60
1.290	0.6840	-0.2849	-2.60
1.295	0.6833	-0.2850	-2.61
1.300	0.6826	-0.2850	-2.62
1.305	0.6821	-0.2850	-2.62
1.310	0.6815	-0.2850	-2.63
1.315	0.6808	-0.2850	-2.64
1.320	0.6803	-0.2851	-2.64
1.325	0.6798	-0.2849	-2.65
1.330	0.6792	-0.2849	-2.66
1.335	0.6785	-0.2849	-2.66
1.340	0.6781	-0.2850	-2.67
1.345	0.6775	-0.2850	-2.67
1.350	0.6769	-0.2848	-2.68
1.355	0.6763	-0.2849	-2.69
1.360	0.6757	-0.2848	-2.69
1.365	0.6751	-0.2847	-2.70
1.370	0.6745	-0.2848	-2.71
1.375	0.6740	-0.2848	-2.71
1.380	0.6733	-0.2846	-2.72
1.385	0.6726	-0.2847	-2.73
1.390	0.6721	-0.2848	-2.73
1.395	0.6714	-0.2847	-2.74
1.400	0.6709	-0.2847	-2.75
1.405	0.6703	-0.2846	-2.75
1.410	0.6696	-0.2846	-2.76
1.415	0.6690	-0.2846	-2.77
1.420	0.6683	-0.2847	-2.78
1.425	0.6676	-0.2847	-2.78
1.430	0.6670	-0.2847	-2.79
1.435	0.6664	-0.2848	-2.80
1.440	0.6657	-0.2847	-2.81
1.445	0.6650	-0.2849	-2.81
1.450	0.6643	-0.2848	-2.82
1.455	0.6636	-0.2848	-2.83
1.460	0.6630	-0.2850	-2.83
1.465	0.6623	-0.2848	-2.84
1.470	0.6616	-0.2849	-2.85
1.475	0.6609	-0.2850	-2.86
1.480	0.6602	-0.2851	-2.86
1.485	0.6594	-0.2851	-2.87
1.490	0.6586	-0.2853	-2.88
1.495	0.6580	-0.2853	-2.89

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.500	0.6572	-0.2853	-2.90
1.505	0.6564	-0.2853	-2.91
1.510	0.6557	-0.2852	-2.91
1.515	0.6549	-0.2854	-2.92
1.520	0.6540	-0.2854	-2.93
1.525	0.6533	-0.2856	-2.94
1.530	0.6525	-0.2855	-2.95
1.535	0.6516	-0.2857	-2.96
1.540	0.6508	-0.2857	-2.97
1.545	0.6501	-0.2857	-2.97
1.550	0.6492	-0.2858	-2.98
1.555	0.6483	-0.2859	-2.99
1.560	0.6476	-0.2858	-3.00
1.565	0.6466	-0.2860	-3.01
1.570	0.6458	-0.2861	-3.02
1.575	0.6448	-0.2862	-3.03
1.580	0.6440	-0.2862	-3.04
1.585	0.6431	-0.2863	-3.05
1.590	0.6423	-0.2865	-3.06
1.595	0.6414	-0.2866	-3.07
1.600	0.6405	-0.2866	-3.08
1.605	0.6398	-0.2868	-3.08
1.610	0.6390	-0.2869	-3.09
1.615	0.6380	-0.2870	-3.10
1.620	0.6372	-0.2870	-3.11
1.625	0.6363	-0.2870	-3.12
1.630	0.6353	-0.2871	-3.13
1.635	0.6346	-0.2873	-3.14
1.640	0.6336	-0.2874	-3.15
1.645	0.6328	-0.2874	-3.16
1.650	0.6317	-0.2876	-3.17
1.655	0.6309	-0.2877	-3.18
1.660	0.6302	-0.2879	-3.19
1.665	0.6291	-0.2878	-3.20
1.670	0.6283	-0.2879	-3.21
1.675	0.6274	-0.2880	-3.22
1.680	0.6266	-0.2882	-3.23
1.685	0.6258	-0.2884	-3.24
1.690	0.6249	-0.2884	-3.24
1.695	0.6242	-0.2884	-3.25
1.700	0.6233	-0.2884	-3.26
1.705	0.6224	-0.2885	-3.27
1.710	0.6217	-0.2886	-3.28
1.715	0.6208	-0.2886	-3.29
1.720	0.6200	-0.2888	-3.30
1.725	0.6191	-0.2887	-3.31

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.730	0.6182	-0.2888	-3.32
1.735	0.6175	-0.2890	-3.33
1.740	0.6166	-0.2889	-3.34
1.745	0.6159	-0.2892	-3.34
1.750	0.6150	-0.2893	-3.35
1.755	0.6142	-0.2892	-3.36
1.760	0.6135	-0.2893	-3.37
1.765	0.6129	-0.2895	-3.38
1.770	0.6121	-0.2897	-3.39
1.775	0.6115	-0.2896	-3.39
1.780	0.6110	-0.2896	-3.40
1.785	0.6103	-0.2895	-3.41
1.790	0.6096	-0.2896	-3.42
1.795	0.6090	-0.2897	-3.42
1.800	0.6082	-0.2896	-3.43
1.805	0.6076	-0.2897	-3.44
1.810	0.6070	-0.2898	-3.44
1.815	0.6064	-0.2899	-3.45
1.820	0.6058	-0.2898	-3.46
1.825	0.6053	-0.2901	-3.46
1.830	0.6047	-0.2900	-3.47
1.835	0.6042	-0.2901	-3.48
1.840	0.6036	-0.2901	-3.48
1.845	0.6032	-0.2901	-3.49
1.850	0.6026	-0.2900	-3.49
1.855	0.6020	-0.2899	-3.50
1.860	0.6015	-0.2898	-3.51
1.865	0.6010	-0.2898	-3.51
1.870	0.6005	-0.2898	-3.52
1.875	0.5999	-0.2899	-3.53
1.880	0.5996	-0.2901	-3.53
1.885	0.5992	-0.2902	-3.53
1.890	0.5990	-0.2901	-3.54
1.895	0.5986	-0.2901	-3.54
1.900	0.5981	-0.2900	-3.55
1.905	0.5977	-0.2901	-3.55
1.910	0.5974	-0.2900	-3.56
1.915	0.5970	-0.2902	-3.56
1.920	0.5966	-0.2901	-3.56
1.925	0.5964	-0.2901	-3.57
1.930	0.5959	-0.2903	-3.57
1.935	0.5957	-0.2903	-3.57
1.940	0.5954	-0.2902	-3.58
1.945	0.5951	-0.2902	-3.58
1.950	0.5948	-0.2903	-3.58
1.955	0.5946	-0.2903	-3.59

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.960	0.5942	-0.2903	-3.59
1.965	0.5939	-0.2902	-3.60
1.970	0.5937	-0.2902	-3.60
1.975	0.5935	-0.2901	-3.60
1.980	0.5932	-0.2901	-3.60
1.985	0.5929	-0.2900	-3.61
1.990	0.5927	-0.2900	-3.61
1.995	0.5924	-0.2900	-3.61
2.000	0.5923	-0.2900	-3.62
2.005	0.5920	-0.2899	-3.62
2.010	0.5917	-0.2899	-3.62
2.015	0.5915	-0.2899	-3.63
2.020	0.5913	-0.2898	-3.63
2.025	0.5911	-0.2899	-3.63
2.030	0.5908	-0.2899	-3.63
2.035	0.5905	-0.2898	-3.64
2.040	0.5902	-0.2898	-3.64
2.045	0.5901	-0.2898	-3.64
2.050	0.5898	-0.2898	-3.65
2.055	0.5895	-0.2896	-3.65
2.060	0.5892	-0.2896	-3.66
2.065	0.5889	-0.2896	-3.66
2.070	0.5887	-0.2897	-3.66
2.075	0.5883	-0.2896	-3.67
2.080	0.5881	-0.2897	-3.67
2.085	0.5877	-0.2897	-3.67
2.090	0.5874	-0.2895	-3.68
2.095	0.5868	-0.2896	-3.68
2.100	0.5866	-0.2897	-3.69
2.105	0.5863	-0.2896	-3.69
2.110	0.5857	-0.2895	-3.70
2.115	0.5854	-0.2898	-3.70
2.120	0.5851	-0.2899	-3.70
2.125	0.5846	-0.2898	-3.71
2.130	0.5841	-0.2899	-3.71
2.135	0.5837	-0.2899	-3.72
2.140	0.5832	-0.2902	-3.72
2.145	0.5828	-0.2901	-3.73
2.150	0.5823	-0.2901	-3.73
2.155	0.5818	-0.2902	-3.74
2.160	0.5813	-0.2901	-3.75
2.165	0.5808	-0.2903	-3.75
2.170	0.5803	-0.2905	-3.76
2.175	0.5797	-0.2905	-3.76
2.180	0.5792	-0.2905	-3.77
2.185	0.5786	-0.2906	-3.78

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.190	0.5781	-0.2907	-3.78
2.195	0.5775	-0.2908	-3.79
2.200	0.5768	-0.2910	-3.79
2.205	0.5763	-0.2910	-3.80
2.210	0.5756	-0.2912	-3.81
2.215	0.5750	-0.2913	-3.81
2.220	0.5744	-0.2913	-3.82
2.225	0.5737	-0.2914	-3.83
2.230	0.5730	-0.2915	-3.84
2.235	0.5724	-0.2916	-3.84
2.240	0.5718	-0.2918	-3.85
2.245	0.5712	-0.2919	-3.86
2.250	0.5705	-0.2920	-3.86
2.255	0.5697	-0.2922	-3.87
2.260	0.5692	-0.2923	-3.88
2.265	0.5685	-0.2924	-3.89
2.270	0.5678	-0.2924	-3.89
2.275	0.5672	-0.2927	-3.90
2.280	0.5664	-0.2929	-3.91
2.285	0.5658	-0.2931	-3.91
2.290	0.5652	-0.2932	-3.92
2.295	0.5645	-0.2932	-3.93
2.300	0.5637	-0.2935	-3.94
2.305	0.5630	-0.2935	-3.95
2.310	0.5623	-0.2937	-3.95
2.315	0.5617	-0.2938	-3.96
2.320	0.5609	-0.2941	-3.97
2.325	0.5602	-0.2942	-3.97
2.330	0.5596	-0.2946	-3.98
2.335	0.5589	-0.2946	-3.99
2.340	0.5582	-0.2949	-4.00
2.345	0.5574	-0.2949	-4.00
2.350	0.5568	-0.2952	-4.01
2.355	0.5562	-0.2954	-4.02
2.360	0.5556	-0.2954	-4.02
2.365	0.5548	-0.2957	-4.03
2.370	0.5540	-0.2957	-4.04
2.375	0.5535	-0.2958	-4.05
2.380	0.5528	-0.2961	-4.05
2.385	0.5522	-0.2963	-4.06
2.390	0.5517	-0.2962	-4.07
2.395	0.5510	-0.2966	-4.07
2.400	0.5504	-0.2967	-4.08
2.405	0.5498	-0.2968	-4.09
2.410	0.5493	-0.2970	-4.09
2.415	0.5488	-0.2972	-4.10

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.420	0.5481	-0.2973	-4.10
2.425	0.5476	-0.2973	-4.11
2.430	0.5472	-0.2975	-4.11
2.435	0.5466	-0.2976	-4.12
2.440	0.5460	-0.2978	-4.13
2.445	0.5455	-0.2979	-4.13
2.450	0.5452	-0.2979	-4.13
2.455	0.5447	-0.2980	-4.14
2.460	0.5442	-0.2981	-4.15
2.465	0.5437	-0.2981	-4.15
2.470	0.5432	-0.2982	-4.16
2.475	0.5428	-0.2983	-4.16
2.480	0.5424	-0.2983	-4.17
2.485	0.5420	-0.2984	-4.17
2.490	0.5416	-0.2984	-4.18
2.495	0.5412	-0.2985	-4.18
2.500	0.5407	-0.2986	-4.18
2.505	0.5404	-0.2987	-4.19
2.510	0.5400	-0.2987	-4.19
2.515	0.5396	-0.2988	-4.20
2.520	0.5392	-0.2989	-4.20
2.525	0.5388	-0.2990	-4.21
2.530	0.5384	-0.2992	-4.21
2.535	0.5381	-0.2992	-4.21
2.540	0.5377	-0.2993	-4.22
2.545	0.5374	-0.2994	-4.22
2.550	0.5371	-0.2994	-4.22
2.555	0.5367	-0.2993	-4.23
2.560	0.5365	-0.2993	-4.23
2.565	0.5364	-0.2990	-4.23
2.570	0.5363	-0.2989	-4.24
2.575	0.5362	-0.2986	-4.24
2.580	0.5363	-0.2983	-4.24
2.585	0.5364	-0.2984	-4.24
2.590	0.5367	-0.2984	-4.24
2.595	0.5367	-0.2987	-4.23
2.600	0.5369	-0.2987	-4.23
2.605	0.5370	-0.2988	-4.23
2.610	0.5370	-0.2990	-4.23
2.615	0.5370	-0.2991	-4.23
2.620	0.5370	-0.2992	-4.23
2.625	0.5371	-0.2994	-4.22
2.630	0.5370	-0.2994	-4.22
2.635	0.5370	-0.2994	-4.22
2.640	0.5371	-0.2995	-4.22
2.645	0.5370	-0.2996	-4.22

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.650	0.5369	-0.2996	-4.22
2.655	0.5368	-0.2997	-4.22
2.660	0.5368	-0.2998	-4.23
2.665	0.5366	-0.2997	-4.23
2.670	0.5364	-0.2999	-4.23
2.675	0.5362	-0.3000	-4.23
2.680	0.5361	-0.3000	-4.23
2.685	0.5361	-0.3000	-4.23
2.690	0.5359	-0.3001	-4.23
2.695	0.5358	-0.3002	-4.23
2.700	0.5358	-0.3003	-4.23
2.705	0.5358	-0.3002	-4.23
2.710	0.5358	-0.3004	-4.23
2.715	0.5358	-0.3003	-4.23
2.720	0.5358	-0.3001	-4.24
2.725	0.5358	-0.3001	-4.23
2.730	0.5359	-0.2999	-4.24
2.735	0.5358	-0.2998	-4.24
2.740	0.5357	-0.2996	-4.24
2.745	0.5356	-0.2994	-4.24
2.750	0.5354	-0.2993	-4.25
2.755	0.5353	-0.2992	-4.25
2.760	0.5351	-0.2991	-4.25
2.765	0.5350	-0.2991	-4.25
2.770	0.5347	-0.2989	-4.26
2.775	0.5345	-0.2988	-4.26
2.780	0.5342	-0.2987	-4.26
2.785	0.5340	-0.2985	-4.27
2.790	0.5338	-0.2985	-4.27
2.795	0.5335	-0.2984	-4.27
2.800	0.5333	-0.2984	-4.28
2.805	0.5331	-0.2983	-4.28
2.810	0.5327	-0.2983	-4.29
2.815	0.5324	-0.2983	-4.29
2.820	0.5322	-0.2982	-4.29
2.825	0.5318	-0.2983	-4.30
2.830	0.5316	-0.2982	-4.30
2.835	0.5313	-0.2982	-4.30
2.840	0.5310	-0.2984	-4.31
2.845	0.5307	-0.2984	-4.31
2.850	0.5304	-0.2984	-4.31
2.855	0.5303	-0.2983	-4.32
2.860	0.5299	-0.2985	-4.32
2.865	0.5296	-0.2984	-4.32
2.870	0.5294	-0.2985	-4.33
2.875	0.5293	-0.2984	-4.33

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.880	0.5291	-0.2981	-4.33
2.885	0.5289	-0.2980	-4.33
2.890	0.5285	-0.2980	-4.34
2.895	0.5281	-0.2977	-4.35
2.900	0.5277	-0.2977	-4.35
2.905	0.5273	-0.2972	-4.36
2.910	0.5268	-0.2973	-4.37
2.915	0.5262	-0.2973	-4.37
2.920	0.5257	-0.2972	-4.38
2.925	0.5252	-0.2972	-4.39
2.930	0.5246	-0.2974	-4.39
2.935	0.5242	-0.2976	-4.40
2.940	0.5238	-0.2975	-4.40
2.945	0.5234	-0.2978	-4.41
2.950	0.5230	-0.2978	-4.41
2.955	0.5225	-0.2979	-4.42
2.960	0.5223	-0.2982	-4.42
2.965	0.5221	-0.2983	-4.42
2.970	0.5218	-0.2985	-4.42
2.975	0.5215	-0.2987	-4.42
2.980	0.5213	-0.2988	-4.42
2.985	0.5212	-0.2989	-4.42
2.990	0.5213	-0.2991	-4.42
2.995	0.5214	-0.2992	-4.42
3.000	0.5212	-0.2993	-4.42
3.005	0.5199	-0.2990	-4.44
3.010	0.5198	-0.2994	-4.44
3.015	0.5199	-0.2995	-4.44
3.020	0.5199	-0.2999	-4.43
3.025	0.5200	-0.3000	-4.43
3.030	0.5201	-0.3001	-4.43
3.035	0.5202	-0.2999	-4.43
3.040	0.5204	-0.2998	-4.43
3.045	0.5205	-0.2995	-4.43
3.050	0.5208	-0.2994	-4.43
3.055	0.5208	-0.2990	-4.43
3.060	0.5207	-0.2988	-4.43
3.065	0.5207	-0.2983	-4.44
3.070	0.5204	-0.2980	-4.44
3.075	0.5203	-0.2976	-4.45
3.080	0.5201	-0.2971	-4.45
3.085	0.5197	-0.2968	-4.46
3.090	0.5192	-0.2967	-4.47
3.095	0.5189	-0.2964	-4.47
3.100	0.5185	-0.2963	-4.48
3.105	0.5180	-0.2963	-4.48

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.110	0.5176	-0.2964	-4.49
3.115	0.5171	-0.2963	-4.50
3.120	0.5167	-0.2964	-4.50
3.125	0.5163	-0.2968	-4.50
3.130	0.5161	-0.2970	-4.50
3.135	0.5159	-0.2974	-4.50
3.140	0.5158	-0.2977	-4.50
3.145	0.5157	-0.2980	-4.50
3.150	0.5155	-0.2984	-4.50
3.155	0.5156	-0.2987	-4.50
3.160	0.5157	-0.2990	-4.49
3.165	0.5159	-0.2995	-4.49
3.170	0.5162	-0.2997	-4.48
3.175	0.5164	-0.2999	-4.48
3.180	0.5167	-0.3002	-4.47
3.185	0.5169	-0.3002	-4.47
3.190	0.5172	-0.2999	-4.47
3.195	0.5176	-0.2998	-4.46
3.200	0.5178	-0.2993	-4.47
3.205	0.5179	-0.2992	-4.46
3.210	0.5179	-0.2988	-4.47
3.215	0.5178	-0.2985	-4.47
3.220	0.5177	-0.2979	-4.48
3.225	0.5176	-0.2977	-4.48
3.230	0.5172	-0.2978	-4.48
3.235	0.5170	-0.2978	-4.49
3.240	0.5167	-0.2980	-4.49
3.245	0.5169	-0.2980	-4.49
3.250	0.5166	-0.2983	-4.49
3.255	0.5166	-0.2984	-4.49
3.260	0.5165	-0.2988	-4.49
3.265	0.5163	-0.2987	-4.49
3.270	0.5166	-0.2992	-4.48
3.275	0.5167	-0.2991	-4.48
3.280	0.5167	-0.2993	-4.48
3.285	0.5166	-0.2995	-4.48
3.290	0.5164	-0.2998	-4.48
3.295	0.5168	-0.3001	-4.47
3.300	0.5168	-0.3000	-4.47
3.305	0.5169	-0.3001	-4.47
3.310	0.5169	-0.3006	-4.47
3.315	0.5169	-0.3007	-4.47
3.320	0.5171	-0.3008	-4.46
3.325	0.5171	-0.3009	-4.46
3.330	0.5173	-0.3010	-4.46
3.335	0.5174	-0.3010	-4.46

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.340	0.5176	-0.3011	-4.45
3.345	0.5176	-0.3011	-4.45
3.350	0.5178	-0.3010	-4.45
3.355	0.5179	-0.3010	-4.45
3.360	0.5179	-0.3010	-4.45
3.365	0.5181	-0.3008	-4.45
3.370	0.5180	-0.3006	-4.45
3.375	0.5180	-0.3005	-4.45
3.380	0.5179	-0.3003	-4.46
3.385	0.5178	-0.3001	-4.46
3.390	0.5177	-0.3000	-4.46
3.395	0.5176	-0.2997	-4.46
3.400	0.5174	-0.2995	-4.47
3.405	0.5173	-0.2992	-4.47
3.410	0.5169	-0.2988	-4.48
3.415	0.5167	-0.2986	-4.48
3.420	0.5164	-0.2985	-4.49
3.425	0.5160	-0.2984	-4.49
3.430	0.5155	-0.2983	-4.50
3.435	0.5152	-0.2980	-4.51
3.440	0.5147	-0.2980	-4.51
3.445	0.5141	-0.2976	-4.52
3.450	0.5136	-0.2976	-4.53
3.455	0.5130	-0.2977	-4.54
3.460	0.5125	-0.2978	-4.54
3.465	0.5121	-0.2980	-4.55
3.470	0.5116	-0.2979	-4.55
3.475	0.5112	-0.2982	-4.56
3.480	0.5107	-0.2984	-4.56
3.485	0.5098	-0.2985	-4.57
3.490	0.5098	-0.2993	-4.57
3.495	0.5095	-0.2995	-4.57
3.500	0.5095	-0.2999	-4.56
3.505	0.5095	-0.3002	-4.56
3.510	0.5095	-0.3004	-4.56
3.515	0.5096	-0.3008	-4.56
3.520	0.5097	-0.3007	-4.56
3.525	0.5102	-0.3009	-4.55
3.530	0.5103	-0.3008	-4.55
3.535	0.5107	-0.3008	-4.54
3.540	0.5115	-0.3000	-4.54
3.545	0.5113	-0.2992	-4.55
3.550	0.5115	-0.2980	-4.55
3.555	0.5115	-0.2970	-4.56
3.560	0.5109	-0.2958	-4.58
3.565	0.5102	-0.2945	-4.60

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.570	0.5090	-0.2935	-4.62
3.575	0.5077	-0.2922	-4.65
3.580	0.5062	-0.2914	-4.67
3.585	0.5038	-0.2908	-4.71
3.590	0.5021	-0.2913	-4.72
3.595	0.4999	-0.2921	-4.75
3.600	0.4987	-0.2932	-4.75
3.605	0.4977	-0.2949	-4.75
3.610	0.4977	-0.2962	-4.74
3.615	0.4976	-0.2974	-4.74
3.620	0.4981	-0.2982	-4.72
3.625	0.4985	-0.2986	-4.71
3.630	0.4992	-0.2994	-4.70
3.635	0.5004	-0.2993	-4.69
3.640	0.5009	-0.2989	-4.68
3.645	0.5011	-0.2984	-4.68
3.650	0.5014	-0.2977	-4.69
3.655	0.5014	-0.2972	-4.69
3.660	0.5014	-0.2967	-4.69
3.665	0.5013	-0.2962	-4.70
3.670	0.5010	-0.2956	-4.71
3.675	0.5012	-0.2952	-4.71
3.680	0.5009	-0.2945	-4.72
3.685	0.5006	-0.2940	-4.72
3.690	0.5001	-0.2932	-4.74
3.695	0.4995	-0.2928	-4.75
3.700	0.4990	-0.2923	-4.76
3.705	0.4981	-0.2917	-4.77
3.710	0.4970	-0.2912	-4.79
3.715	0.4960	-0.2910	-4.81
3.720	0.4946	-0.2911	-4.82
3.725	0.4933	-0.2916	-4.84
3.730	0.4913	-0.2922	-4.86
3.735	0.4901	-0.2935	-4.86
3.740	0.4893	-0.2952	-4.86
3.745	0.4888	-0.2968	-4.85
3.750	0.4894	-0.2987	-4.83
3.755	0.4905	-0.3003	-4.80
3.760	0.4921	-0.3014	-4.78
3.765	0.4941	-0.3016	-4.75
3.770	0.4962	-0.3011	-4.73
3.775	0.4978	-0.3005	-4.71
3.780	0.4987	-0.2977	-4.72
3.785	0.4979	-0.2960	-4.74
3.790	0.4971	-0.2945	-4.76
3.795	0.4964	-0.2938	-4.78

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.800	0.4953	-0.2932	-4.80
3.805	0.4939	-0.2929	-4.82
3.810	0.4932	-0.2931	-4.83
3.815	0.4921	-0.2933	-4.84
3.820	0.4914	-0.2934	-4.85
3.825	0.4906	-0.2937	-4.85
3.830	0.4899	-0.2943	-4.86
3.835	0.4899	-0.2963	-4.84
3.840	0.4897	-0.2965	-4.84
3.845	0.4897	-0.2967	-4.84
3.850	0.4895	-0.2967	-4.85
3.855	0.4893	-0.2971	-4.85
3.860	0.4891	-0.2973	-4.85
3.865	0.4889	-0.2972	-4.85
3.870	0.4889	-0.2974	-4.85
3.875	0.4887	-0.2975	-4.85
3.880	0.4885	-0.2976	-4.85
3.885	0.4883	-0.2977	-4.85
3.890	0.4879	-0.2981	-4.85
3.895	0.4878	-0.2980	-4.86
3.900	0.4875	-0.2983	-4.86
3.905	0.4872	-0.2986	-4.86
3.910	0.4868	-0.2986	-4.87
3.915	0.4864	-0.2988	-4.87
3.920	0.4862	-0.2991	-4.87
3.925	0.4858	-0.2994	-4.87
3.930	0.4856	-0.2995	-4.88
3.935	0.4850	-0.2995	-4.88
3.940	0.4848	-0.2998	-4.88
3.945	0.4843	-0.3000	-4.89
3.950	0.4839	-0.3003	-4.89
3.955	0.4834	-0.3004	-4.90
3.960	0.4831	-0.3006	-4.90
3.965	0.4826	-0.3006	-4.90
3.970	0.4821	-0.3009	-4.91
3.975	0.4817	-0.3010	-4.91
3.980	0.4812	-0.3013	-4.92
3.985	0.4805	-0.3015	-4.93
3.990	0.4801	-0.3017	-4.93
3.995	0.4794	-0.3019	-4.94
4.000	0.4789	-0.3022	-4.94
4.005	0.4784	-0.3023	-4.95
4.010	0.4776	-0.3025	-4.95
4.015	0.4772	-0.3026	-4.96
4.020	0.4765	-0.3027	-4.97
4.025	0.4760	-0.3028	-4.97

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
4.030	0.4754	-0.3031	-4.98
4.035	0.4746	-0.3032	-4.99
4.040	0.4741	-0.3035	-4.99
4.045	0.4735	-0.3035	-5.00
4.050	0.4729	-0.3036	-5.01
4.055	0.4720	-0.3038	-5.02
4.060	0.4714	-0.3041	-5.02
4.065	0.4707	-0.3041	-5.03
4.070	0.4699	-0.3044	-5.04
4.075	0.4692	-0.3046	-5.05
4.080	0.4685	-0.3047	-5.05
4.085	0.4680	-0.3048	-5.06
4.090	0.4673	-0.3050	-5.07
4.095	0.4667	-0.3054	-5.07
4.100	0.4658	-0.3053	-5.08
4.105	0.4648	-0.3056	-5.09
4.110	0.4644	-0.3059	-5.10
4.115	0.4635	-0.3061	-5.11
4.120	0.4627	-0.3064	-5.11
4.125	0.4620	-0.3065	-5.12
4.130	0.4611	-0.3069	-5.13
4.135	0.4601	-0.3070	-5.14
4.140	0.4592	-0.3073	-5.15
4.145	0.4585	-0.3074	-5.16
4.150	0.4574	-0.3076	-5.17
4.155	0.4566	-0.3077	-5.18
4.160	0.4558	-0.3077	-5.19
4.165	0.4553	-0.3079	-5.20
4.170	0.4544	-0.3077	-5.21
4.175	0.4537	-0.3075	-5.22
4.180	0.4531	-0.3074	-5.23
4.185	0.4527	-0.3076	-5.24
4.190	0.4521	-0.3076	-5.24
4.195	0.4515	-0.3077	-5.25
4.200	0.4512	-0.3077	-5.25
4.205	0.4508	-0.3080	-5.26
4.210	0.4504	-0.3080	-5.26
4.215	0.4499	-0.3080	-5.27
4.220	0.4497	-0.3082	-5.27
4.225	0.4490	-0.3084	-5.28
4.230	0.4488	-0.3086	-5.28
4.235	0.4484	-0.3087	-5.28
4.240	0.4480	-0.3088	-5.29
4.245	0.4478	-0.3090	-5.29
4.250	0.4475	-0.3093	-5.29
4.255	0.4472	-0.3093	-5.29

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
4.260	0.4470	-0.3095	-5.29
4.265	0.4467	-0.3098	-5.29
4.270	0.4465	-0.3098	-5.30
4.275	0.4464	-0.3100	-5.30
4.280	0.4462	-0.3102	-5.30
4.285	0.4461	-0.3102	-5.30
4.290	0.4459	-0.3104	-5.30
4.295	0.4457	-0.3105	-5.30
4.300	0.4456	-0.3106	-5.30
4.305	0.4454	-0.3108	-5.30
4.310	0.4454	-0.3108	-5.30
4.315	0.4451	-0.3109	-5.31
4.320	0.4449	-0.3110	-5.31
4.325	0.4447	-0.3112	-5.31
4.330	0.4445	-0.3113	-5.31
4.335	0.4443	-0.3115	-5.31
4.340	0.4441	-0.3118	-5.31
4.345	0.4441	-0.3120	-5.31
4.350	0.4440	-0.3124	-5.31
4.355	0.4439	-0.3127	-5.31
4.360	0.4438	-0.3129	-5.30
4.365	0.4440	-0.3132	-5.30
4.370	0.4440	-0.3135	-5.30
4.375	0.4441	-0.3138	-5.29
4.380	0.4443	-0.3141	-5.29
4.385	0.4442	-0.3143	-5.28
4.390	0.4447	-0.3147	-5.28
4.395	0.4450	-0.3149	-5.27
4.400	0.4453	-0.3151	-5.26
4.405	0.4457	-0.3151	-5.26
4.410	0.4457	-0.3152	-5.26
4.415	0.4462	-0.3152	-5.25
4.420	0.4463	-0.3155	-5.25
4.425	0.4465	-0.3155	-5.24
4.430	0.4468	-0.3154	-5.24
4.435	0.4468	-0.3154	-5.24
4.440	0.4469	-0.3153	-5.24
4.445	0.4470	-0.3154	-5.24
4.450	0.4468	-0.3154	-5.24
4.455	0.4470	-0.3156	-5.24
4.460	0.4471	-0.3151	-5.24
4.465	0.4474	-0.3152	-5.24
4.470	0.4477	-0.3152	-5.23
4.475	0.4481	-0.3153	-5.23
4.480	0.4485	-0.3156	-5.22
4.485	0.4489	-0.3160	-5.21

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
4.490	0.4494	-0.3163	-5.20
4.495	0.4499	-0.3168	-5.19
4.500	0.4506	-0.3172	-5.18
4.505	0.4511	-0.3174	-5.17
4.510	0.4517	-0.3178	-5.16
4.515	0.4521	-0.3180	-5.15
4.520	0.4527	-0.3180	-5.14
4.525	0.4532	-0.3182	-5.13
4.530	0.4535	-0.3182	-5.13
4.535	0.4538	-0.3182	-5.13
4.540	0.4541	-0.3180	-5.12
4.545	0.4541	-0.3181	-5.12
4.550	0.4544	-0.3180	-5.12
4.555	0.4546	-0.3180	-5.12
4.560	0.4548	-0.3179	-5.12
4.565	0.4550	-0.3179	-5.11
4.570	0.4551	-0.3179	-5.11
4.575	0.4552	-0.3178	-5.11
4.580	0.4553	-0.3178	-5.11
4.585	0.4554	-0.3177	-5.11
4.590	0.4554	-0.3177	-5.11
4.595	0.4554	-0.3176	-5.11
4.600	0.4553	-0.3175	-5.11
4.605	0.4554	-0.3174	-5.11
4.610	0.4552	-0.3174	-5.11
4.615	0.4552	-0.3174	-5.12
4.620	0.4550	-0.3174	-5.12
4.625	0.4549	-0.3174	-5.12
4.630	0.4549	-0.3173	-5.12
4.635	0.4546	-0.3173	-5.12
4.640	0.4545	-0.3173	-5.13
4.645	0.4545	-0.3172	-5.13
4.650	0.4543	-0.3173	-5.13
4.655	0.4543	-0.3171	-5.13
4.660	0.4543	-0.3172	-5.13
4.665	0.4539	-0.3170	-5.14
4.670	0.4543	-0.3171	-5.13
4.675	0.4543	-0.3171	-5.13
4.680	0.4543	-0.3168	-5.13
4.685	0.4543	-0.3169	-5.13
4.690	0.4540	-0.3169	-5.13
4.695	0.4542	-0.3169	-5.13
4.700	0.4544	-0.3166	-5.13
4.705	0.4540	-0.3165	-5.14
4.710	0.4542	-0.3163	-5.14
4.715	0.4541	-0.3162	-5.14

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.720	0.4541	-0.3158	-5.14
4.725	0.4538	-0.3156	-5.15
4.730	0.4537	-0.3154	-5.15
4.735	0.4535	-0.3151	-5.16
4.740	0.4531	-0.3149	-5.17
4.745	0.4524	-0.3148	-5.17
4.750	0.4521	-0.3148	-5.18
4.755	0.4516	-0.3147	-5.19
4.760	0.4510	-0.3147	-5.19
4.765	0.4506	-0.3148	-5.20
4.770	0.4499	-0.3149	-5.21
4.775	0.4498	-0.3150	-5.21
4.780	0.4493	-0.3151	-5.21
4.785	0.4494	-0.3154	-5.21
4.790	0.4486	-0.3153	-5.22
4.795	0.4487	-0.3150	-5.22
4.800	0.4486	-0.3149	-5.22
4.805	0.4486	-0.3150	-5.22
4.810	0.4484	-0.3147	-5.23
4.815	0.4484	-0.3143	-5.23
4.820	0.4484	-0.3137	-5.24
4.825	0.4483	-0.3132	-5.24
4.830	0.4482	-0.3130	-5.24
4.835	0.4483	-0.3126	-5.25
4.840	0.4484	-0.3123	-5.25
4.845	0.4483	-0.3121	-5.25
4.850	0.4482	-0.3117	-5.26
4.855	0.4485	-0.3117	-5.25
4.860	0.4482	-0.3116	-5.26
4.865	0.4481	-0.3115	-5.26
4.870	0.4480	-0.3115	-5.26
4.875	0.4480	-0.3116	-5.26
4.880	0.4480	-0.3117	-5.26
4.885	0.4479	-0.3119	-5.26
4.890	0.4478	-0.3120	-5.26
4.895	0.4477	-0.3121	-5.26
4.900	0.4476	-0.3122	-5.26
4.905	0.4476	-0.3124	-5.26
4.910	0.4476	-0.3126	-5.26
4.915	0.4476	-0.3127	-5.26
4.920	0.4475	-0.3129	-5.26
4.925	0.4475	-0.3130	-5.25
4.930	0.4475	-0.3131	-5.25
4.935	0.4475	-0.3134	-5.25
4.940	0.4474	-0.3134	-5.25
4.945	0.4473	-0.3135	-5.25

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.950	0.4476	-0.3138	-5.25
4.955	0.4476	-0.3140	-5.24
4.960	0.4476	-0.3141	-5.24
4.965	0.4478	-0.3144	-5.24
4.970	0.4479	-0.3144	-5.24
4.975	0.4480	-0.3146	-5.23
4.980	0.4481	-0.3149	-5.23
4.985	0.4480	-0.3151	-5.23
4.990	0.4483	-0.3153	-5.22
4.995	0.4486	-0.3155	-5.22
5.000	0.4488	-0.3158	-5.21
5.005	0.4490	-0.3163	-5.21
5.010	0.4492	-0.3165	-5.20
5.015	0.4495	-0.3167	-5.19
5.020	0.4495	-0.3172	-5.19
5.025	0.4499	-0.3176	-5.18
5.030	0.4500	-0.3179	-5.18
5.035	0.4503	-0.3185	-5.17
5.040	0.4504	-0.3187	-5.17
5.045	0.4505	-0.3191	-5.16
5.050	0.4508	-0.3192	-5.16
5.055	0.4508	-0.3195	-5.15
5.060	0.4508	-0.3196	-5.15
5.065	0.4508	-0.3198	-5.15
5.070	0.4509	-0.3197	-5.15
5.075	0.4506	-0.3196	-5.15
5.080	0.4507	-0.3194	-5.15
5.085	0.4507	-0.3193	-5.16
5.090	0.4509	-0.3189	-5.16
5.095	0.4507	-0.3188	-5.16
5.100	0.4510	-0.3184	-5.16
5.105	0.4509	-0.3183	-5.16
5.110	0.4511	-0.3181	-5.16
5.115	0.4511	-0.3179	-5.16
5.120	0.4515	-0.3177	-5.16
5.125	0.4516	-0.3178	-5.16
5.130	0.4519	-0.3176	-5.16
5.135	0.4521	-0.3174	-5.15
5.140	0.4522	-0.3174	-5.15
5.145	0.4525	-0.3173	-5.15
5.150	0.4527	-0.3174	-5.15
5.155	0.4527	-0.3173	-5.15
5.160	0.4528	-0.3173	-5.15
5.165	0.4529	-0.3173	-5.15
5.170	0.4529	-0.3173	-5.15
5.175	0.4530	-0.3174	-5.14

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
5.180	0.4529	-0.3173	-5.15
5.185	0.4528	-0.3173	-5.15
5.190	0.4528	-0.3172	-5.15
5.195	0.4525	-0.3171	-5.15
5.200	0.4524	-0.3170	-5.16
5.205	0.4522	-0.3171	-5.16
5.210	0.4520	-0.3171	-5.16
5.215	0.4518	-0.3171	-5.16
5.220	0.4516	-0.3170	-5.17
5.225	0.4514	-0.3169	-5.17
5.230	0.4508	-0.3167	-5.18
5.235	0.4506	-0.3166	-5.18
5.240	0.4503	-0.3164	-5.19
5.245	0.4498	-0.3163	-5.20
5.250	0.4494	-0.3161	-5.20
5.255	0.4490	-0.3158	-5.21
5.260	0.4486	-0.3158	-5.22
5.265	0.4481	-0.3156	-5.22
5.270	0.4475	-0.3155	-5.23
5.275	0.4469	-0.3153	-5.24
5.280	0.4464	-0.3151	-5.25
5.285	0.4457	-0.3150	-5.26
5.290	0.4452	-0.3149	-5.27
5.295	0.4448	-0.3151	-5.27
5.300	0.4442	-0.3147	-5.28
5.305	0.4436	-0.3149	-5.29
5.310	0.4432	-0.3147	-5.30
5.315	0.4426	-0.3147	-5.30
5.320	0.4423	-0.3146	-5.31
5.325	0.4417	-0.3144	-5.32
5.330	0.4412	-0.3146	-5.32
5.335	0.4407	-0.3145	-5.33
5.340	0.4402	-0.3143	-5.34
5.345	0.4398	-0.3143	-5.34
5.350	0.4394	-0.3142	-5.35
5.355	0.4389	-0.3137	-5.36
5.360	0.4384	-0.3135	-5.37
5.365	0.4381	-0.3136	-5.37
5.370	0.4374	-0.3131	-5.38
5.375	0.4370	-0.3128	-5.39
5.380	0.4361	-0.3123	-5.41
5.385	0.4353	-0.3121	-5.42
5.390	0.4345	-0.3121	-5.43
5.395	0.4337	-0.3121	-5.44
5.400	0.4332	-0.3119	-5.45
5.405	0.4321	-0.3123	-5.46

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.410	0.4316	-0.3120	-5.47
5.415	0.4308	-0.3121	-5.48
5.420	0.4303	-0.3119	-5.49
5.425	0.4299	-0.3118	-5.50
5.430	0.4289	-0.3119	-5.51
5.435	0.4285	-0.3121	-5.51
5.440	0.4278	-0.3120	-5.52
5.445	0.4271	-0.3122	-5.53
5.450	0.4268	-0.3124	-5.53
5.455	0.4264	-0.3124	-5.54
5.460	0.4258	-0.3123	-5.55
5.465	0.4254	-0.3122	-5.55
5.470	0.4249	-0.3122	-5.56
5.475	0.4242	-0.3120	-5.57
5.480	0.4238	-0.3120	-5.58
5.485	0.4233	-0.3119	-5.58
5.490	0.4226	-0.3117	-5.59
5.495	0.4220	-0.3116	-5.60
5.500	0.4212	-0.3116	-5.61
5.505	0.4206	-0.3115	-5.62
5.510	0.4200	-0.3116	-5.63
5.515	0.4194	-0.3117	-5.64
5.520	0.4187	-0.3119	-5.64
5.525	0.4183	-0.3121	-5.65
5.530	0.4179	-0.3123	-5.65
5.535	0.4174	-0.3125	-5.66
5.540	0.4170	-0.3127	-5.66
5.545	0.4166	-0.3129	-5.66
5.550	0.4162	-0.3130	-5.67
5.555	0.4158	-0.3131	-5.67
5.560	0.4155	-0.3132	-5.67
5.565	0.4153	-0.3134	-5.68
5.570	0.4149	-0.3134	-5.68
5.575	0.4144	-0.3137	-5.68
5.580	0.4139	-0.3137	-5.69
5.585	0.4136	-0.3138	-5.69
5.590	0.4130	-0.3141	-5.70
5.595	0.4124	-0.3144	-5.70
5.600	0.4118	-0.3146	-5.71
5.605	0.4113	-0.3151	-5.71
5.610	0.4108	-0.3154	-5.71
5.615	0.4104	-0.3160	-5.71
5.620	0.4098	-0.3165	-5.72
5.625	0.4096	-0.3172	-5.71
5.630	0.4093	-0.3177	-5.71
5.635	0.4089	-0.3182	-5.71

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.640	0.4086	-0.3187	-5.71
5.645	0.4086	-0.3194	-5.70
5.650	0.4085	-0.3196	-5.70
5.655	0.4083	-0.3200	-5.70
5.660	0.4081	-0.3205	-5.70
5.665	0.4079	-0.3209	-5.70
5.670	0.4078	-0.3214	-5.69
5.675	0.4077	-0.3218	-5.69
5.680	0.4073	-0.3226	-5.69
5.685	0.4072	-0.3228	-5.69
5.690	0.4069	-0.3233	-5.68
5.695	0.4071	-0.3236	-5.68
5.700	0.4068	-0.3241	-5.68
5.705	0.4067	-0.3244	-5.68
5.710	0.4066	-0.3250	-5.67
5.715	0.4066	-0.3253	-5.67
5.720	0.4066	-0.3256	-5.67
5.725	0.4066	-0.3260	-5.66
5.730	0.4066	-0.3266	-5.65
5.735	0.4065	-0.3268	-5.65
5.740	0.4064	-0.3271	-5.65
5.745	0.4064	-0.3274	-5.65
5.750	0.4064	-0.3278	-5.64
5.755	0.4064	-0.3280	-5.64
5.760	0.4064	-0.3284	-5.64
5.765	0.4064	-0.3287	-5.64
5.770	0.4064	-0.3289	-5.63
5.775	0.4063	-0.3291	-5.63
5.780	0.4062	-0.3291	-5.63
5.785	0.4062	-0.3293	-5.63
5.790	0.4061	-0.3293	-5.63
5.795	0.4059	-0.3295	-5.63
5.800	0.4059	-0.3295	-5.63
5.805	0.4057	-0.3296	-5.63
5.810	0.4057	-0.3295	-5.64
5.815	0.4055	-0.3296	-5.64
5.820	0.4052	-0.3295	-5.64
5.825	0.4050	-0.3295	-5.65
5.830	0.4048	-0.3295	-5.65
5.835	0.4046	-0.3295	-5.65
5.840	0.4044	-0.3293	-5.65
5.845	0.4042	-0.3292	-5.66
5.850	0.4037	-0.3291	-5.67
5.855	0.4035	-0.3292	-5.67
5.860	0.4031	-0.3289	-5.68
5.865	0.4026	-0.3290	-5.68

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
5.870	0.4021	-0.3286	-5.69
5.875	0.4017	-0.3286	-5.70
5.880	0.4012	-0.3285	-5.70
5.885	0.4008	-0.3286	-5.71
5.890	0.4003	-0.3284	-5.72
5.895	0.4001	-0.3283	-5.72
5.900	0.3995	-0.3282	-5.73
5.905	0.3992	-0.3283	-5.73
5.910	0.3984	-0.3284	-5.74
5.915	0.3983	-0.3283	-5.74
5.920	0.3980	-0.3282	-5.75
5.925	0.3972	-0.3280	-5.76
5.930	0.3971	-0.3277	-5.77
5.935	0.3965	-0.3275	-5.78
5.940	0.3962	-0.3274	-5.78
5.945	0.3961	-0.3274	-5.78
5.950	0.3953	-0.3268	-5.80
5.955	0.3947	-0.3265	-5.81
5.960	0.3944	-0.3262	-5.82
5.965	0.3935	-0.3258	-5.83
5.970	0.3927	-0.3258	-5.84
5.975	0.3920	-0.3256	-5.86
5.980	0.3911	-0.3255	-5.87
5.985	0.3903	-0.3259	-5.87
5.990	0.3894	-0.3257	-5.89
5.995	0.3889	-0.3260	-5.89
6.000	0.3882	-0.3259	-5.90

2 Measurement Quality

Reliable band: from 45 MHz (S11 error σ = 1.0e-03); the points below it are kept in the plot (shaded, outside the scale) and in the tables, but are not reliable.
Spike removal: on (isolated point, 2nd difference > 6 x local noise and > S11 error budget, not shared with another sweep); replaced 1 point(s): Short at 4885 MHz ($|\Delta S11|$ = 1.4e-03).
Outliers shared by several sweeps, left alone (instrument artefacts cancelled by the calibration): 3835 MHz.

- Unreliable below 45 MHz: 8 point(s) (3 mm probe, S11 error σ = 1.0e-03). There the reference liquids resolve ε_r only to ±17.2 at 5 MHz … ±2.1 at 40 MHz per σ (usable: $\leq$ ±2). Values kept, but left out of the chart scale and of the summary medians; start the sweep at $\geq$ 45 MHz.

- Gn shows no radiation signature even at the top of the sweep (3 GHz – 6 GHz): mean arg(Gn) = -108° (radiation: -90°, median deviation 20°) and $|Gn| \propto f^{0.4}$ (radiation: f^3). Over the whole sweep it holds noise and the inconsistency between the reference liquids instead (its term is ≈7% of the Ref 1 (Distilled Water) admittance), so the full method's radiation correction is empirical here: compare with the simplified cross-check, especially for samples whose ε_r lies outside the range of the reference liquids.

- 2 frequency point(s) excluded: reference liquids nearly indistinguishable ($|S11_ref1 - S11_ref2| < 1e\text{-}02$). Consider starting the sweep above the affected range.